Indira Gandhi National Open University
School of Computer and Information
Sciences (SOCIS)

MCS-062
Introduction to Data Science

Block

2

INTRODUCTION TO DATA SCIENCE-2

Unit 5 Data Preparation for Analysis

Unit 6 Descriptive and Exploratory Data Analysis – An Introduction

Unit 7 Inferential and Predictive Data Analysis – An Introduction

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BLOCK 2 INTRODUCTION TO DATA SCIENCE-2

Block-2 of the course "Introduction to Data Science" focuses on one of the most critical phases of the data science lifecycle—data preparation and analysis. This block serves as the bridge between raw data and actionable insights, equipping learners with the skills needed to preprocess data effectively and apply analytical techniques that inform decisions in both academic research and industry operations.

Unit 5: Data Preprocessing addresses the foundational steps necessary to prepare data for meaningful analysis. It covers essential tasks such as data cleaning, integration, transformation, and reduction. The unit also introduces concepts like data curation and knowledge discovery, which are vital when dealing with real-world datasets that are often incomplete or inconsistent. Mastery of these skills is indispensable for data scientists and analysts working in fields ranging from business intelligence to public health and scientific research.

Unit 6: Descriptive and Exploratory Data Analysis – An Introduction introduces statistical techniques used to summarize and explore data patterns. Learners gain exposure to various data types, sampling techniques, and methods for descriptive and exploratory analysis. These techniques are routinely used in industries to understand customer behavior, operational performance, and market trends, making them highly applicable in practical settings.

Unit 7: Inferential and Predictive Data Analysis – An Introduction extends the analytical scope by focusing on statistical inference and prediction. Topics include hypothesis testing, confidence intervals, and basic predictive modeling. These tools are crucial in making data-driven forecasts and decisions in sectors such as finance, logistics, healthcare, and policymaking. The emphasis on predictive methods aligns with the growing demand for professionals who can derive forward-looking insights from historical data.

Overall, Block-2 empowers learners to transform raw data into meaningful interpretations and forecasts. By combining theoretical knowledge with practical techniques, this block prepares students for advanced data science topics and ensures their readiness to tackle real-world data challenges in both academic and professional environments.

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UNIT 5 DATA PREPARATION FOR ANALYSIS

Structure

5.0 Introduction
5.1 Objectives
5.2 Need for Data Preparation
5.3 Data Preprocessing
5.3.1 Data Cleaning
5.3.2 Data Integration
5.3.3 Data Reduction
5.3.4 Data Transformation
5.4 Selection and Data Extraction
5.5 Data Curation
5.5.1 Steps of Data Curation
5.5.2 Importance of Data Curation
5.6 Data Integration
5.6.1 Data Integration Techniques
5.6.2 Data Integration Approaches
5.7 Knowledge Discovery
5.8 Summary
5.9 Solutions/Answers
5.10 Further Readings

5.0 INTRODUCTION

Having established a strong foundation in the fundamentals of data science, types of data, and data acquisition tools and techniques through Block-1 (Units 1 to 4), learners are now prepared to move into the next crucial phase of the data science pipeline: preparing data for analysis. Block-1 equipped students with the knowledge to identify, classify, and collect data from diverse sources using appropriate tools and methodologies. However, real-world data is often messy, incomplete, inconsistent, or redundant—posing significant challenges before any meaningful analysis can be conducted.

Unit 5, "Data Preprocessing," builds directly upon this foundation by introducing the essential techniques required to transform raw data into a structured, clean, and analysis-ready format. It emphasizes the importance of data cleaning, integration, transformation, and reduction—key processes that ensure data quality and consistency. These tasks are not just technical necessities but critical enablers of trustworthy analysis and reliable decision-making.

The unit also introduces the concept of data curation and its significance in maintaining the integrity and usability of datasets over time. By learning how

to curate and prepare data, students develop practical skills that bridge the gap between data collection (explored in Block-1) and deeper statistical or machine learning tasks that follow in later units.

This progression from data understanding and acquisition to data preprocessing reflects the real-world workflow of a data scientist or analyst. It reinforces the idea that high-quality input data is the backbone of accurate and insightful analytics, making Unit 5 a natural and essential continuation of the learning journey that began in Block-1.

5.1 OBJECTIVES

After finishing this unit, you will be to be able to:

- Describe the meaning of "data quality".
- Explain basic techniques for data preprocessing.
- Use the technique of data selection and extraction.
- Define data curation and data integration.
- Describe the knowledge discovery.

5.2 NEED FOR DATA PREPARATION

In the present time, data is one of the key resources for a business. Any raw facts are known as data and are processed to create information; information is integrated to create knowledge. Since knowledge is power, it has evolved into a modern currency, which is valued and traded between parties. Everyone wants to discuss the knowledge and benefits they can gain from it. Data is the most significant resource available to marketers, agencies, publishers, media firms, and others. But only high-quality data is useful. A good quality data set is suitable for decision-making. How do you gauge the data quality? The usefulness of the data for the intended purpose and its completeness, accuracy, timeliness, consistency, validity, and uniqueness are used to determine the data's quality.

In simpler terms, data quality refers to how accurate and helpful the data are for the task at hand. Further, data quality also relates to the actions that apply the necessary quality management procedures and methodologies to make sure the data is useful and actionable for the data consumers. A wide range of elements, including accuracy, completeness, consistency, timeliness, uniqueness, and validity, influence data quality. Figure 1 shows the basic factors of data quality.

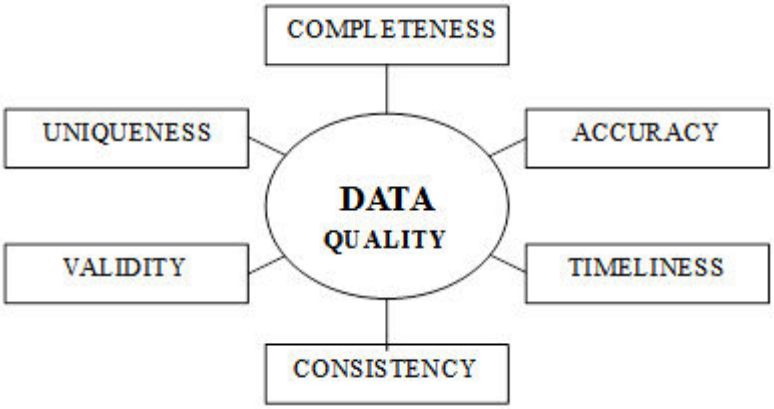

Figure 5.1: Factors of Data Quality

These factors are explained below:

- **Accuracy** - The data must be true and reflect events that take place in the real world. Accuracy measures determine how closely the figures agree with the verified right information sources.
- **Completeness** - The degree to which the data is complete determines how well it can provide the necessary value.
- **Consistency** - Data consistency is the homogeneity of the data across applications, networks. Data consistency is most desirable when the data is gathered from several sources. For example, identical datasets should not conflict if they are stored in different locations.
- **Timeliness** - Data that is timely is readily available whenever it is needed. The timeliness factor also entails keeping the data accurate to make sure it is always available, accessible, and updated in real time.
- **Uniqueness** - Uniqueness is defined as the lack of duplicate or redundant data across all datasets. The collection should contain zero duplicate records.
- **Validity** - Data must be obtained in compliance with the firm's defined business policies and guidelines. The data should adhere to the appropriate and standard formats. In addition, all the dataset values should be within the specified ranges.

Consider yourself a manager at a company, say XYZ Pvt Ltd, who has been asked to research the sales statistics for a specific organization, say ABC. You immediately get to work on this project by carefully going through the ABC company's database and data warehouse for the parameters or dimensions (such as the products, prices, and units sold etc.), which may be used in your study. However, your enthusiasm suffers a major problem when you see that several of the attributes for different tuples do not have any recorded values. You want to incorporate the information in your study on whether each item purchased was marked in the related purchase records, but you find that this data has not been recorded. According to users of this database system, the data recorded for some transactions contained mistakes; therefore, such transactions were causing anomalies in the data. Thus, this data was not registered.

The three characteristics of data quality—accuracy, completeness, and consistency—are highlighted in the paragraph above. Large databases and data warehouses used in the real world frequently contain inaccurate, incomplete, and inconsistent data. What may be the causes of such erroneous data in the databases? There may be problems with the data collection tools, which may result in mistakes during the data-entry processes. Some of these problems can be due to the following reasons:

- Personal biases: There can be instances when users do not want to submit their personal information. They may purposefully enter inaccurate data values for required fields in such cases. For example, if a user is asked to enter his/her date of birth, a person may select the default date as his/her birthdate (please note in many applications, the default date may be "January 1"). This inaccurate data is referred to as *disguised missing data*.
- Data transfer errors: Data transfer errors may be generated during data communication due to certain technical limitations. For example, the use of synchronized data transit and consumption may be constrained by technological limitations, such as a short buffer capacity.
- The unreliable data may result from differences in naming conventions, data codes, or input field types (e.g., the format of date fields such as dd/mm/yy or mm/dd/yyyy) etc.
- Finally, cleansing of data may be needed to remove duplicate tuples.

Incomplete data can be caused by a variety of circumstances. Certain properties, such as customer information for sales transaction data, might not always be accessible. It is likely that some data might be omitted since it was not thought to be important at the time of input. A misinterpretation or malfunctioning technology could prevent the recording of essential data. For example, the data, which did not match the previously stored data, may be eliminated. Furthermore, it is likely that the data's past alterations or histories are not documented. For tuples with missing values for some properties, it could be required to infer missing data.

5.3 DATA PREPROCESSING

Preprocessing is the process of taking raw data and turning it into information that may be used for data analysis. Data cleaning, data integration, data reduction and data transformation are the main phases of data preprocessing (see Figure 2). In addition, data discretization is another component of data preprocessing. You may refer to the further readings for more details on data discretization.

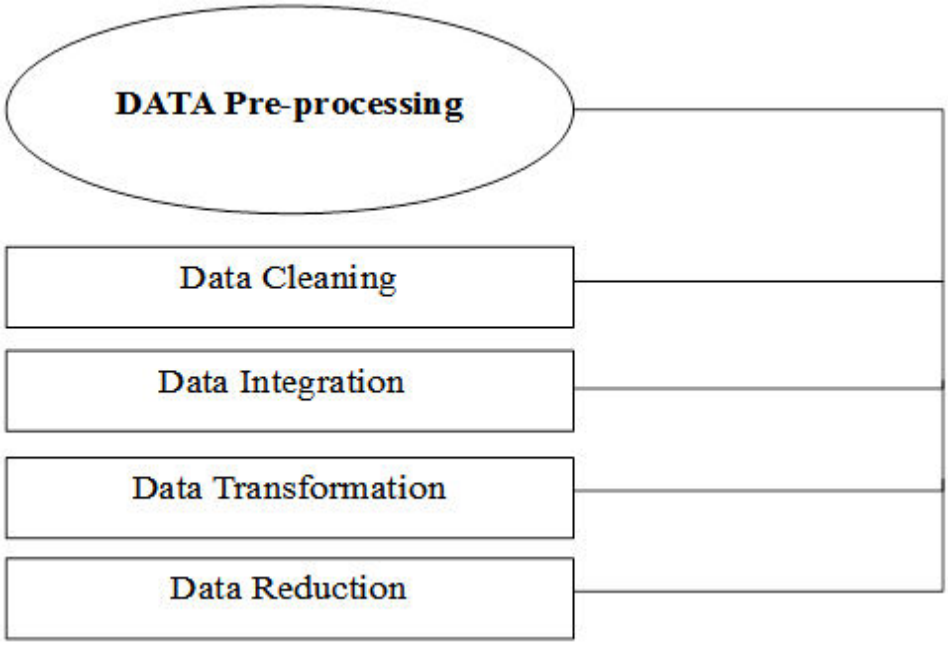

Figure 5.2: Data preprocessing

5.3.1 Data Cleaning

Data cleaning is an essential step in data preprocessing. It is also referred to as scrubbing. It is crucial for the construction of a good analysis model. Data cleaning is a required but frequently overlooked aspect of data preprocessing. Real-world data typically exhibit incompleteness, noise, and inconsistency. In addition to addressing discrepancies, this task entails filling in missing numbers, smoothing out noisy data, and eliminating outliers. Generally, a good data cleaning process helps reduce errors in data modeling and enhances data quality. Although it might be a time-consuming and laborious operation, it is necessary to fix data inaccuracies and delete bad entries. In the subsequent paragraphs, we discuss some basic data cleansing operations.

a. **Missing Values**

Consider you need to study *Customer and Sales* data for ABC Company. You examined the data and pointed out that numerous tuples lack recorded values for several characteristics or attributes (for example, customer income). The following techniques can be used to add values that are lacking.

i) **Ignore the tuple**: Consider that a data science task involves classification based on the attribute(s) whose value is/are missing. In such a case, ignoring the tuples with missing classifying attribute(s) values would be desirable. This method is particularly detrimental when each attribute has a significantly different percentage of missing values. In such a case, we disregard the remaining characteristics or attributes in the tuple.

ii) **Manually enter the omitted value**: In general, this strategy is time-consuming and might not be practical for huge data sets with a substantial number of missing values.

iii) **Fill up the blank with a global constant**: A single constant, such as "Unknown" or "−∞", should be used to replace all missing attribute values. If missing data are replaced with "Unknown", the analysis algorithm can mistakenly think that all "Unknown" values collectively comprise a valid data category. So, despite being simple, this strategy is not perfect.

iv) **To fill in the missing value, use a measure of the attribute's central tendency (such as the mean or median)**: The median should be used for skewed data distributions, whereas the mean can be used for normal (symmetric) data distribution. Assume, for instance, that the ABC company's customer income data distribution is symmetric and that the mean income is INR 50,000/-. Use this value to fill in the income value that is missing.

v) **For all samples that belong to the same class as the specified tuple, use the mean or median**: For instance, if we were to categorize customers based on their credit risk, the mean income value of customers who belonged to the same credit risk category as the given tuple, might be used to fill in the missing value. If the data distribution is skewed for the relevant class, it is best to utilize the median value.

vi) **Fill in the blank with the value that is most likely to be there**: This result can be reached using regression, inference-based techniques using a Bayesian formalization, or decision tree induction. For example, using the other characteristics of your customer's data, you may create a decision tree to forecast the missing income of a specific customer.

b. **Noisy Data**

Noise is the variance or random error in a measured variable. Noise can be an outlier, which can be recognized by employing tools for data visualization and basic statistical description techniques such as scatter plots and boxplots. How can the data be "smoothed" out to reduce noise if the given data has numeric property (for example, the price of the items)? The following are some of the data-smoothing strategies.

i) **Binning:** Binning techniques smooth sorted data values by looking at their "neighborhood" or nearby values. The values, which have been sorted, are divided into various "buckets" or bins. Binning techniques carry out local smoothing since they look at the data values' surroundings. Smoothing by bin means that each value in a bin is changed to the mean value of that bin. As an illustration, suppose a bin contains three numbers 5, 10 and 12. The average of these three numbers in the bin is 9. Consequently, the value nine replaces each of the bin's original values.

Similarly, smoothing by bin medians, which substitutes the bin median for each bin value, can be used. In addition, you can use the maximum or the minimum values in a bin, also referred to as bin boundaries of a specific bin, in place of bin mean or median. This type of smoothing is called smoothing by bin boundaries. This

method uses the nearest boundary value to replace each bin value. In general, the smoothing effect increases with increasing breadth of bin boundaries. The bins can also be created using identical intervals, leading to constant ranges of values.

ii) **Regression:** Regression is a method of mapping the data values to a function. This regression function may then be used to smooth out the data. Finding the "best" line to fit two traits (or variables) is the goal of linear regression, which enables one attribute to predict the other. As an extension of linear regression, multiple linear regression involves more than two features and fits the data to a multidimensional surface.

iii) **Outlier analysis**: Clustering groups comparable values into "clusters". These clusters can then be used to identify outliers. It makes sense to classify values that are outliers as being outside the set of clusters.

iv) **Data discretization:** Data discretization is a data transformation and reduction technique. It is an extensively used data smoothing technique. Data discretization decreases the number of distinct values for each property using some technique; for instance, you can discretize.

v) **Data using the binning approach**: Concept hierarchies are a data discretization technique that can also be applied to smooth out the data. The quantity of data values that the analysis process must process is decreased by a concept hierarchy. For example, the price variable, which represents the price value of commodities, may be discretized into "lowly priced", "moderately priced", and "expensive" categories.

Steps of Data Cleaning

Data cleaning is a critical step in the data preparation process that helps ensure data quality and accuracy. While specific data cleaning steps may vary depending on the nature of the data and the organization's needs, here are some basic steps to create a framework for data cleaning.

1. Remove duplicate or irrelevant observations: Duplicate data may be produced when data sets from various sources are combined, scraped, or data is obtained from clients or other departments.

2. Fix structural errors: When measuring or transferring data, you may come across structural mistakes such as unusual naming practices, typographical errors, or wrong capitalization. Such inconsistencies may lead to mislabeled categories or classes. For instance, "N/A" and "Not Applicable", which might be present on any given document, may create two different classifications. Instead, they should be considered under the same heading or should be considered as missing values.

3. Managing unwanted outliers: Outliers can cause the formation of inaccurate models; for example, outliers can result in the construction of an incorrect regression function. However, decision tree models are more

robust to outliers than linear regression. We should not eliminate outliers unless there is a compelling reason to do so. Sometimes removing them can improve performance, but not always. Therefore, the outlier must be eliminated for a good cause, such as suspicious measurements that are unlikely to be present in real-life data.

4. Handling missing data: Missing data is a deceptively difficult issue in machine learning. We cannot just ignore or remove the missing observations. They must be carefully treated since they can indicate a serious problem. Data gaps resemble puzzle pieces that are missing. Dropping it is equivalent to denying that the puzzle slot is there. It is like trying to put a piece from another puzzle into this one. Furthermore, you need to be aware of how you report missing data. Instead of just filling it in with the mean, you can effectively let the computer choose the appropriate constant to account for missingness by using this flagging and filling method.

5. Validate and QA: You should be able to respond to these inquiries as part of fundamental validation following the data cleansing process, for example:
   - Does the data make sense?
   - Does the data abide by the regulations that apply to its particular field?
   - Does it support or refute your hypothesis? Does it offer any new information?
   - Can you see patterns in the data that will support your analysis?
   - Is there a problem with the data quality?

Methods of Data Cleaning

The following are some of the methods of data cleaning. Some of the methods have already been discussed as steps of data cleaning.

1. **Ignore the tuples:** This approach is not particularly practical because it can only be used when a tuple has multiple characteristics and missing values.

2. **Fill in the missing value:** This strategy is neither practical nor very effective. Additionally, the process could take a long time. The missing value must be entered in the approach. The most common method for doing this is manually putting the value for the missing value. Other options include using the attribute mean or the value with the highest probability for the missing value.

3. **Binning method:** The Binning method, as explained earlier, creates bins of nearby values. These values, then, are used to smooth the data.

4. **Regression:** The regression function can predict and smooth the data for machine learning. You may use bivariate or multivariate linear regression for this purpose.

5. **Clustering:** The primary objective of the clustering approach is to group the comparable values in a cluster, thus, creating data clusters. Clustering also aids in ascertaining possible outliers.

5.3.2 Data Integration

Data from many sources, such as files, data cubes, databases (both relational and non-relational), etc., may be combined before a machine learning algorithm can use it as training or test data. The data from the sources may have the following characteristics:

- The data sources may be homogeneous or heterogeneous.
- The data sources may contain structured, unstructured, or semi-structured data.

Redundancies and inconsistencies can be reduced and avoided with careful integration. The following are some of the issues of data integration.

a. **Entity identification problem:** In many projects, data from several sources are integrated into a consistent data set. For example, a data warehouse gathers data from several sources into coherent data storage. These sources include various databases, data cubes, and flat files.

   During data integration, there are several things to consider. Integration of schemas and object matching might be challenging. How is it possible to match comparable real-world things across different data sources? The entity identification problem describes this. How can a computer or data analyst be sure that a client's ID in one database and their customer number in another database refer to the same attribute? Examples of metadata for each attribute include the name of the attribute, its definition, data type, permissible range of values and null rules for handling empty, zero, or null values. Such metadata can be used to avoid errors during the integration of the schema. The data may also be transformed with the aid of metadata. For example, in two different instances of data of an organization, the code for pay data might be "H" for high income and "S" for small income in one instance of the database; the same pay code in another instance of a database may be 1 and 2.

   When comparing attributes from one database to another during integration, the data structure must be considered properly. This is done to make sure that any referential constraints and functional dependencies on attributes present in the source system are also present in the target system. For instance, one system might apply a discount to the entire order but only to certain items by another. If this is not found before integration, things in the target system can be incorrect.

b. **Redundancy and Correlation Analysis:** Another crucial problem in data integration is redundancy. If an attribute (like annual income, for example) can be "derived" from another attribute or group of data, it may be redundant. Inconsistent attributes or dimension names can also bring redundancies in the final data set.

Correlation analysis can identify some redundancies. Based on the available data, such analysis can quantify the strength of relationships between two attributes. We employ the chi-square test ($\chi 2$) for finding relationships between nominal data. Numeric attributes can be analyzed using the correlation coefficient and covariance, which look at how one attribute's values differ from those of another.

c. **Tuple duplication:** Duplication should be identified at the tuple level, e.g., when, for a particular unique data entry case, there are two or more identical tuples. Additional data redundant sources include – deformalized tables, which are frequently used to increase performance by reducing joins; faulty data entry or updating some (not all) redundant data occurrences, etc. Inconsistencies frequently appear between different duplicates. For example, there may be inconsistency as the same purchaser's name may appear with multiple addresses within the purchase order database. This might happen if a database for purchase orders has attributes for the buyer's name and address rather than a foreign key to this data.

d. **Data value conflict detection and resolution:** Data value conflicts must be found and resolved as part of data integration. As an illustration, attribute values from many sources may vary for the same real-world thing. Variations in representation, scale, or encoding may be the cause of this. In one system, a weight attribute might be maintained in British imperial units, while in another, metric units. For a hotel chain, the cost of rooms in several cities could include various currencies, services (such as a complimentary breakfast) and taxes. Similarly, every university may have its own curriculum and grading system. When sharing information among them, one university might use the quarter system, provide three database systems courses, and grade students from A+ to F, while another would use the semester system, provide two database systems courses, and grade students from 1 to 10. Information interchange between two such universities is challenging to establish accurate course-to-grade transformation procedures. An attribute in one system might be recorded at a lower abstraction level than the "identical" attribute in another since the abstraction level of attributes might also differ. As an illustration, an attribute with the same name in one database may relate to the total sales of one branch of a company, however, the same attribute name in another database can refer to the company's overall regional shop sales.

5.3.3 Data Reduction

In this phase, data is trimmed. The number of records, attributes, or dimensions can be reduced. When reducing data, one should keep in mind that the outcomes from the reduced data should be identical to those from the original data. Consider that you have chosen some data for analysis from *ABC Company's* data warehouse. The data set will probably be enormous! Large-scale complex data analysis and mining can be time-consuming, rendering such a study impractical or unfeasible. Techniques for data reduction can be applied to create a condensed version of the data set that is

considerably smaller while meticulously retaining the integrity of the original data. In other words, mining the smaller data set should yield more useful results while effectively yielding the same analytical outcomes. This section begins with an overview of data reduction tactics and then delves deeper into specific procedures. Data compression, dimensionality reduction, and numerosity reduction are data reduction methods.

a. **Dimensionality reduction** refers to the process of lowering the number of random variables or qualities. Principal components analysis and wavelet transformations are techniques used to reduce data dimensions by transforming or rescaling the original data. By identifying and eliminating duplicated, weakly relevant, or irrelevant features or dimensions, attribute subset selection is a technique for dimensionality reduction.

b. **Numerosity reduction** strategies substitute different, more compact forms of data representation for the original data volume. Both parametric and non-parametric approaches are available. In parametric techniques, a model is employed to estimate the data, which frequently necessitates the maintenance of only the data parameters rather than the actual data. (Outliers may also be stored.) Examples include log-linear models and regression. Non-parametric methods include the use of histograms, clustering, sampling, and data cube aggregation to store condensed versions of the data.

c. **Transformations** are used in **data compression** to create a condensed or "compressed" version of the original data. Data compression can be - lossless data compression, in which the original data can be recovered from the compressed data without losing any information, or lossy data compression, in which only a fraction of the original data can be recovered from the compressed data on decompression. There are a number of lossless string compression algorithms; however, they typically permit only a small amount of data manipulation. Techniques for reducing numerosity and dimensions can also be categorized as data compression methods.

There are other additional data reduction techniques. You may please note that the data reduction techniques also require computational time. The time saved by analysis on a smaller data set should not be "erased" or outweighed by the computational effort required for data reduction.

Data Discretization: - It is regarded as a component of data reduction. The data discretization alters numerical data by converting values to intervals or concept labels. These techniques enable data analysis at various levels of granularity by automatically generating concept hierarchies for the data. Binding, histogram analysis, decision tree analysis, cluster analysis, and correlation analysis are examples of discretization techniques. Concept hierarchies for nominal data may be produced based on the definitions of the schema and the distinct attribute values for every attribute.

5.3.4 Data Transformation

This procedure is used to change the data into formats that are suited for the analytical process. Data transformation involves transforming or consolidating the data into analysis-ready formats. The following are some data transformation strategies:

a. **Smoothing**: Smoothing is a process which attempts to reduce data noise. You can use methods like binning, regression, and grouping for data smoothening.

b. **Attribute construction** (or *feature construction*): Attribute construction is the process of constructing additional attributes using the set of data attributes. The primary objective of attribute construction is to aid the analysis process,

c. **Aggregation:** Aggregation is the process of summarizing the data of an attribute based on some criteria; for instance, the daily sales data may be combined to produce monthly or yearly sales. This process is often used to build a data cube for data analysis at different levels of abstraction.

d. **Normalizations**: Normalization resizes attribute data to fit a narrower range: −1.0 to 1.0; or 0.0 to 1.0.

e. **Discretization**: Discretization is the process where interval labels replace the raw values of a numeric attribute (e.g., age) (e.g., 0–10, 11–20, etc.) or conceptual labels (e.g., *youth, adult*, *senior*). A concept hierarchy for the number attribute can then be created by recursively organizing the labels into higher-level concepts. To meet the demands of different users, more than one concept hierarchy might be built for the same attribute.

f. **Concept hierarchy creation using nominal data:** These concept hierarchies allow for extrapolating higher-level concepts, for example - a street to concepts like a city or a country. At the schema definition level, numerous hierarchies for nominal qualities can be automatically created and are implicit in the database structure.

Check Your Progress 1:

1. What is meant by data preprocessing?
2. Why is preprocessing important?
3. What are the 5 characteristics of data processing?
4. What are the major steps of data preprocessing?
5. What is data cleaning?
6. What is the importance of data cleaning?
7. What are the main steps of Data Cleaning?

5.4 SELECTION AND DATA EXTRACTION

The process of choosing the best data source, data type, and collection tools is known as data selection. Prior to starting the actual data collection procedure, data selection is conducted. This concept makes a distinction between selective data reporting (excluding data that is not supportive of a study premise) and active/interactive data selection (using obtained data for monitoring activities/events or conducting secondary data analysis). Data integrity may be impacted by how acceptable data are selected for a research project.

The main goal of data selection is to choose the proper data type, source, and tool that enables researchers to solve research issues effectively. This decision typically depends on the research area, research questions, prior research, and the availability of the data sources.

When cost and convenience considerations make you decide which "appropriate" data to collect, then you may face data integrity issues. Cost and convenience are unquestionably important variables to consider while deciding. However, researchers should consider how much these factors can skew the results of their study.

Data Selection Issues

When choosing data, researchers should be conscious of a few things, including:

- Researchers should select the right type and appropriate data sources to respond appropriately to the stated research questions viz, appropriate methods for obtaining representative samples and appropriate tools for gathering data. It is difficult to separate the choice of data type and source from the tools used to get the data. The type/source of data and the procedures used to collect the data should be compatible.

Types and Sources of data: Different data sources and data types can be expressed in a variety of ways. The following are the two main categories of data types:

- Quantitative data are expressed as numerical measurements at the interval and ratio levels.
- Qualitative data can take the form of text, images, music, and video.

Although preferences within scientific disciplines differ as to which type of data is preferred, some researchers employ information from both quantitative and qualitative sources to comprehend a certain event more thoroughly.

Researchers get data from people that may be qualitative (such as by studying child-rearing techniques) or quantitative (biochemical recording markers and anthropometric measurements). Field notes, journals, laboratory notes, specimens, and first-hand observations of people, animals, and plants can all be used as data sources.

Choosing the right data is discipline-specific and primarily influenced by the purpose of investigation, the body of prior research, and the availability of data sources. The following list of questions will help you choose the right data type and sources:

- What is the research question?
- What is the investigation's field of study? (This establishes the guidelines for any investigation. Selected data should not go beyond what is necessary for the analysis.)
- According to the literature (prior research), what kind of information should be gathered?
- Which form of data—qualitative, quantitative, or a combination of both—should be considered?

Data extraction is the process of gathering or obtaining many forms of data from numerous sources, many of which may be erratically organised or wholly unstructured. Consolidating, processing and refining data enables information to be kept in a centralized area so that it can be altered. These data storage units could be local, online, or a combination of the two.

Data Extraction and ETL

To appreciate the importance of data extraction, let us briefly define the ETL process.

1. **Extraction:** One or more sources or systems are used to collect the data. During the extraction phase, relevant data is located, identified, and then prepared for processing or transformation. One can finally analyse them for business knowledge by combining various data types through extraction.
2. **Transformation**: Once the data has been effectively extracted, it can be refined further. Data is cleansed, sorted, and structured as part of the transformation process. Duplicate entries will be removed, missing values will be filled in or removed, and audits will be performed to offer reliable, consistent, and usable data.
3. **Loading**: The high-quality data, which is generated by the transformation process, is subsequently sent to a single, centralized target location for storage and analysis.

Data extraction tools

The tools listed below can be used for tasks other than simple data extraction. The features of these software are given below:

1. **Scrape storm:** Scrape storm is software that uses AI to scrape the web or gather data. It is compatible with Windows, Mac, or Linux operating systems and has a simple and straightforward visual operation. This program automatically detects objects like emails, numbers, lists, forms, links, photos, and prices. When transferring extracted data to Excel, MongoDB, CSV, HTML, TXT, MySQL, SQL Server, PostgreSQL, Google Sheets, or WordPress, it can make use of a variety of export strategies.

2. **Altair Monarch**: Monarch is a desktop-based self-service and does not require any programming. It can link to various data sources, including big data, cloud-based data, and both organised and unstructured data. It connects to, cleans, and processes data with high speed and no errors. It employs more than 80 built-in data preparation functions. When you use this tool, you waste less time making data legible; therefore, you can spend more time creating higher-level knowledge from the data.
3. **Klippa:** Klippa can process contracts, invoices, receipts, and passports over the cloud. For most documents, the conversion time taken by the software may be between one and five seconds. The data classification and manipulation may be done online, round-the-clock. It supports a variety of file types, including PDF, JPG, and PNG. It can also convert between JSON, PDF/A, XLSX, CSV, and XML. Additionally, the software handles file sharing, custom branding, payment processing, cost management, and invoicing management.
4. NodeXL: Basic version of the software is a free, open-source add-on extension for Microsoft Excel 2007, 2010, 2013, and 2016. Since the software is an add-on, data integration is not performed; instead, it focuses on social network analytics. Advanced network analytics, text and sentiment analysis, and robust report generating are extra capabilities included with NodeXL Pro.

Check Your Progress 2:

1. What is the data selection process?

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2. What is Data Extraction and define the term ETL?

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3. What are the challenges of data extraction?

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5.5 DATA CURATION

In the previous sections, we have discussed data preprocessing. The objective of data preprocessing is to obtain high-quality data for analysis. In this section, we discuss data curation. Data curation is integrating and maintaining high-quality data for a specific purpose in an organization using multiple data sources. Data curation aims to create high-quality data sets that can be accessed and used by data users, who may include company

employees or any other person looking for such information. Data curation involves collecting the data from its sources; integrating and arranging the collected data; indexing the information so generated; and categorizing the information to support business decisions, academic needs, scientific research, etc. Data curation is a step in the more extensive data management process that helps prepare data sets for usage in business intelligence (BI) and analytics applications.

5.5.1 Steps of Data Curation

There are numerous tasks involved in curating data sets, which can be divided into the following main steps.

- The data that will be required for the proposed analytics applications should be determined.
- Map the data sets and note the metadata that goes with them.
- Collect the data sets.
- The data should be ingested into a system, such as, a data lake, a data warehouse, etc.
- Clean the data to remove abnormalities, inconsistencies, and mistakes, including missing values, duplicate records, and spelling mistakes.
- Model, organize, and transform the data to prepare it for specific analytics applications.
- To make the data sets accessible to users, create searchable indexes.
- Maintain and manage the data in compliance with the requirements of continuous analytics and the laws governing data privacy and security.

5.5.2 Importance of Data Curation

The following are the reasons for performing data curation.

1. **Helps to organize pre-existing data for a corporation**: Businesses produce a large amount of data on a regular basis. For example, when a customer clicks on a website, adds something to their cart, or completes a transaction, an online retailer might record that information. Data curators assist businesses in better understanding vast amounts of information by integrating and aggregating the data into data sets.
2. **Connects professionals in different departments**: When a company engages in data curation, it often brings together people from several departments who might not typically collaborate. Data curators might collaborate with stakeholders, system designers, data scientists, and data analysts to collect and transfer information.
3. **High-quality data:** It typically uses organizational techniques that make it simple to grasp and have fewer errors. Curators may make sure that a company's research and information continue to be of the highest caliber because the data curation process entails cleansing the data. Removing unnecessary information may facilitate better data set structure.
4. **Makes data easy to understand:** Data curators make sure there are no

errors and utilize proper formatting. This makes it simpler for specialists, who are not knowledgeable about a research issue, to comprehend a data set.

5. **Allows for higher cost and time efficiency**: A business may spend more time and money organizing and distributing data if it does not regularly employ data curation. Because prior data is already organized and distributed, businesses that routinely do data curation may be able to save time, effort, and money. Businesses can reduce the time to obtain and process data by using data curators.

Check Your Progress 3:

1. What is the Importance of Data Curation?

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2. Explain Data Curation.

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3. What are the goals of data curation?

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4. What are the benefits of data curation?

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5.6 DATA INTEGRATION

Data Integration creates coherent data storage by combining data from several sources. Smooth data integration is facilitated by the resolution of semantic heterogeneity, metadata, correlation analysis, tuple duplicate identification, and data conflict detection. It is a tactic that combines data from several sources so that data consumers may access it in a single, consistent view that displays the required data. Systems can communicate using flat files, data cubes, or numerous databases. Data integration is crucial because it maintains data accuracy while providing a consistent view of dispersed data. It helps the analysis tools extract valuable information, which in turn helps the executive and management make tactical choices that will benefit the company.

5.6.1 Data Integration Techniques

Manual Integration-When integrating data, this technique avoids employing automation. The data analyst gathers, purifies, and integrates the data to create actionable data. A small business with a modest amount of data can use this approach. Nevertheless, manual data integration, which is extensive, complex, and ongoing, will take a lot of time.

Middleware Integration-Data from many sources are combined, normalized, and stored in the final data set using middleware software. This method is employed when an organization integrates data from historical systems into modern systems. Software called middleware serves as a translator between outdated and contemporary systems. You may be required to connect the interfaces of the two systems. Such integration is designed for specific systems only.

Application-based Integration- It uses software applications to extract, transform, and load data from various sources. Although this strategy saves time and effort, it is a little more difficult because creating such an application requires technical knowledge.

Uniform Access Integration- This method integrates information from a wider range of sources. In this instance, however, the data is left in its initial place and is not moved. To put it simply, this technique produces a unified view of the combined data. The integrated data does not need to be saved separately because the end user only sees the integrated view.

5.6.2 Data Integration Approaches

There are two basic data integration approaches. These are –

Tight Coupling- It combines data from many sources into a single physical location using ETL (Extraction, Transformation, and Loading) tools.

Loose Coupling- The source databases are the most efficient place to store facts with loose coupling. This method offers an interface that receives a user query, converts it into a format that the source databases can understand, and immediately transmits the question to the source databases to get the answer.

Check Your Progress 4:

1. What is meant by data integration?

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2. What is an example of data integration?

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3. What is the purpose of data integration?

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5.7 KNOWLEDGE DISCOVERY

Knowledge discovery in databases is the process of obtaining pertinent knowledge from a body of data (KDD). This well-known knowledge discovery method includes several processes, such as data preparation and selection, data cleansing, incorporating prior knowledge about the data sets, and interpreting precise answers from the observed results.

Marketing, fraud detection, telecommunications, and manufacturing are some of the key KDD application areas. In the last ten years, the KDD process has reached its pinnacle. Inductive learning, Bayesian statistics, semantic query optimisation, knowledge acquisition for expert systems, and information theory are just a few of the numerous discovery-related methodologies. The ultimate objective of KDD is to extract high-level knowledge from low-level data. Due to the accessibility and quantity of data available today, knowledge discovery is a challenge of astounding importance and necessity. Given how swiftly the topic has expanded, it is not surprising that professionals and experts today have access to a variety of treatments of data.

Steps of Knowledge Discovery

1. **Developing an understanding of the application domain:** Knowledge discovery starts with this preliminary step. It establishes the framework for selecting the best course of action for a variety of options, such as transformation, algorithms, etc. The individuals in charge of a KDD project need to be aware of the end users' goals as well as the environment in which the knowledge discovery process will take place.

2. **Selecting and producing the data set that will be used for discovery:** Once the objectives have been specified, the data that will be used for the knowledge discovery process should be identified. Some of the basic functions for this step are - determining what data is accessible, identifying what information is expected to be obtained from the data, and combining all the data for knowledge discovery into one data set, etc. Knowledge discovery is important since it extracts knowledge and insight from the given data. This provides the framework for building the models.

3. **Preprocessing and cleansing:** This step helps in increasing the data reliability. For instance, the outcome of the Data Mining supervised approach may be considered not dependable if it is established that a specific attribute is unreliable or has a sizable amount of missing data. Preprocessing comprises data cleaning, which includes handling the missing quantities and removing noise or outliers. Such data cleansing might make use of sophisticated statistical methods or analysis algorithms. After developing a prediction model for these features, missing data can be forecasted.

4. **Data Transformation:** This phase entails creating and preparing the necessary data for knowledge discovery. Here, techniques of attribute transformation (such as discretization of numerical attributes, functional transformation etc.) and dimensionality reduction (such as feature selection, feature extraction, record sampling etc.) are employed. This step, in general, is different for every project. However, this step is essential for the success of the KDD project, as a proper transformation results in proper analysis and proper conclusions.

5. **Prediction and description:** Once data transformation is complete, you can decide to analyze data using classification, regression, clustering, or any other method based on KDD objectives. A forecast and a description are the two main objectives of knowledge discovery. The visualization aspects are included in descriptive knowledge discovery. Most knowledge discovery techniques use inductive learning, which generalizes enough prepared models to produce a model either explicitly or implicitly. The fundamental premise of the inductive technique is that the trained model holds for the examples that follow.

6. **Choosing a Knowledge Discovery Algorithm:** After determining the approach, the next step involves selecting appropriate strategies. At this stage, a specific technique needs to be applied to uncover patterns. When comparing precision and understandability, neural networks enhance the former, while decision trees improve the latter. There are multiple ways in which each meta-learning system could achieve success. The objective of meta-learning is to elucidate why a data analysis algorithm succeeds or fails in solving a specific problem. Consequently, this methodology aims to comprehend the conditions under which a data analysis algorithm is most efficient. Each algorithm encompasses parameters and learning techniques, such as tenfold cross-validation or alternative divisions for training and testing.

7. **Implementing the Data Analysis Algorithm**: Ultimately, the data analysis algorithm is deployed. The approach might require multiple iterations before yielding a desirable outcome. For example, by varying the algorithms, you can modify variables such as the minimum number of instances in a single decision tree leaf.

8. **Evaluation:** In this phase, the patterns, principles, and reliability of the outcomes from the knowledge discovery process are evaluated and interpreted in the context of the objective established in the previous step. At this point, we examine the preprocessing steps and their influence on the results. For instance, you might introduce an additional feature in step 4 and then proceed. The key focal points in this stage are the comprehensibility and effectiveness of the generated model. Additionally, any identified knowledge is documented during this stage for future reference.

Check Your Progress 5:

1. What is Knowledge Discovery?

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2. What are the Steps involved in Knowledge Discovery?

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3. What are knowledge discovery tools?

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4. Explain the process of KDD.

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5.8 SUMMARY

Despite the development of several methods for preparing data, the intricacy of the issue and the vast amount of inconsistent or unclean data mean that this field of study is still very active. This unit gives a general overview of data preprocessing and describes how to turn raw data into usable information. The preprocessing of the raw data included data integration, data reduction, transformation, and discretization. In this unit, we have discussed five different data-cleaning techniques that can make data more reliable and produce high-quality results. Building, organizing, and maintaining data sets is known as data curation. A data curator usually determines the necessary data sets and makes sure they are gathered, cleaned up, and changed as necessary. The curator is also in charge of providing users with access to the data sets and information related to them, such as their metadata and lineage documentation. The data curator's primary goal is to ensure users have access to the appropriate data for analysis and decision-making. Data integration is the procedure of fusing information from diverse sources into a single, coherent data store. The unit also introduced knowledge discovery techniques and procedures.

5.9 SOLUTIONS/ANSWERS

Check Your Progress 1:

1. As a part of data preparation, data preprocessing refers to any type of processing done on raw data to get it ready for a data processing technique. It has long been regarded as a crucial first stage in data mining and machine learning.
2. It raises the reliability and accuracy of the data. Preprocessing data can increase the correctness and quality of a dataset, making it more dependable by removing missing or inconsistent data values brought by human or computer mistakes. It ensures consistency in data.
3. Data quality is characterized by five characteristics: correctness, completeness, reliability, relevance, and timeliness.
4. The five major steps of data preprocessing are:
   - Data quality assessment
   - Data cleaning
   - Data transformation
   - Data reduction
5. The practice of correcting or deleting inaccurate, damaged, improperly formatted, duplicate, or incomplete data from a dataset is known as data cleaning. There are numerous ways for data to be duplicated or incorrectly categorised when merging multiple data sources.
6. Data cleansing, sometimes referred to as data cleaning or scrubbing, is the process of locating and eliminating mistakes, duplication, and irrelevant data from a raw dataset. Data cleansing, which is a step in the preparation of data, ensures that the cleaned data is used to create accurate, tenable visualizations, models, and business choices.
7. Step 1: Remove irrelevant data; Step 2: Deduplicate your data; Step 3: Fix structural errors; Step 4: Deal with missing data; Step 5: Filter out data outliers; Step 6: Validate your data.

Check Your Progress 2:

1. The process of retrieving data from the database that are pertinent to the analysis activity is known as data selection. Sometimes the data selection process comes before data transformation and consolidation.
2. Data extraction is the process of gathering or obtaining many forms of data from numerous sources, many of which may be erratically organised or wholly unstructured. The process of extracting, transforming, and loading data is called ETL. Thus, ETL integrates information from several data sources into a single consistent data store, which can be a data warehouse or data analytics system.
3. The cost and time involved in extracting data, as well as the accuracy of the data, are obstacles. The correctness of the data depends on the quality of the data source, which can be an expensive and time-consuming procedure.

Check Your Progress 3:

1. It entails gathering, organizing, indexing, and cataloging information for users within an organization, a group, or the wider public. Data curation can help with academic needs, commercial decisions, scientific research, and other endeavors.
2. The process of producing, arranging, and managing data sets so that users who are looking for information can access and use them is known as data curation. Data must be gathered, organized, indexed, and cataloged for users within an organization, group, or the broader public.
3. By gathering pertinent data into organized, searchable data assets, data curation's overarching goal is to speed up the process of extracting insights from raw data.
4. The benefits of data curation are:
   - Easily discover and use data.
   - Ensure data quality.
   - Maintain metadata linked with data.
   - Ensure compliance through data lineage and classification.

Check Your Progress 4:

1. Data integration is used to bring together data from several sources to give people a single perspective. Making data more readily available, easier to consume, and easier to use by systems and users is the foundation of data integration.
2. In the case of customer data integration, information about each customer is extracted from several business systems, such as sales, accounts, and marketing, and combined into a single picture of the client for use in customer service, reporting, and analysis.
3. Data integration combines data collected from various platforms to increase its value for your company. It enables your staff to collaborate more effectively and provide more for your clients. You cannot access the data collected in different systems without data integration.

Check Your Progress 5:

1. Knowledge discovery is the labor-intensive process of extracting implicit, unknown-before information from databases that may be beneficial.
2. Steps of Knowledge Discovery:
   - Developing an understanding of the application domain
   - Selecting and producing the data set that will be used for the discovery.
   - Preprocessing and cleansing
   - Data Transformation
   - Prediction and description
   - Deciding on a data analysis algorithm

- Utilizing the data analysis algorithm
- Evaluation

3. The process can benefit from a variety of qualitative and quantitative methods and techniques, such as knowledge surveys, questionnaires, one-on-one and group interviews, focus groups, network analysis, and observation. It can be used to locate communities and specialists.
4. Knowledge Discovery from Data, often known as KDD, is another commonly used phrase that is treated as a synonym for data mining. Others see data mining as just a crucial stage in the knowledge discovery process when intelligent techniques are used to extract data patterns. The steps involved in knowledge discovery from data are as follows:
   - Data cleaning (to remove noise or irrelevant data).
   - Data integration (where multiple data sources may be combined).
   - Data selection (where data relevant to the analysis task is retrieved from the database).
   - Data transformation (where data is consolidated into forms appropriate for mining by performing summary or aggregation functions, for sample).
   - Data mining (an important process where intelligent methods are applied to extract data patterns).
   - Pattern evaluation (to identify the fascinating patterns representing knowledge based on some interestingness measures).
   - Knowledge presentation (where knowledge representation and visualization techniques are used to present the mined knowledge to the user).

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UNIT 6 DESCRIPTIVE AND EXPLORATORY DATA ANALYSIS – AN INTRODUCTION

Structure

6.0 Introduction

6.1 Objectives

6.2 Data Science - Definition

6.3 Types of Data

6.3.1 Statistical Data Types

6.3.2 Sampling

6.4 Basic Methods of Data Analysis

6.4.1 Descriptive Analysis

6.4.2 Exploratory Analysis

6.5 Summary

6.6 Solutions/Answers

6.0 INTRODUCTION

Building on the foundational understanding of data science principles (Unit 1), data types and structures (Unit 2), data acquisition methods (Units 3 and 4), and data preprocessing techniques (Unit 5), this Unit i.e. Unit 6 marks a significant transition into the analytical phase of the data science workflow. Once data has been collected and cleaned, the next logical step is to begin exploring and summarizing it to uncover patterns, trends, and initial insights.

Unit 6, "Descriptive and Exploratory Data Analysis – An Introduction," provides the learner with tools and techniques to perform preliminary analysis that reveals the structure and behavior of the data. This unit emphasizes the role of descriptive statistics—such as measures of central tendency, variability, and distribution—and introduces exploratory methods that help in understanding data relationships without formal modeling. These techniques act as a diagnostic toolkit, allowing analysts to ask meaningful questions and guide the selection of further statistical or predictive methods.

The knowledge gained from Block-1 enables students to recognize the importance of data characteristics and quality, while Unit 5 ensures that the data used in this unit is well-prepared for meaningful analysis. Together, they set the stage for Unit 6, where learners begin to derive initial interpretations, generate hypotheses, and visualize data behaviour—critical capabilities in both academic research and real-world analytics projects.

By the end of this unit, learners will be equipped not just with analytical techniques, but with the ability to think critically about what the data is suggesting. This unit thus acts as a bridge between data preparation and more advanced analytical reasoning, ensuring a smooth and logical progression in the data science learning pathway.

This Unit introduces you to the basic concept of data sciences. This Unit provides an introduction to different types of data used in data science. It also points to other kinds of analysis that can be performed using data science. Further, the Unit also introduces some of the common mistakes of data science.

6.1 OBJECTIVES

At the end of this Unit, you should be able to:

- Define the term data science in the context of an organisation
- Explain different types of data
- Explain Descriptive analysis
- Explain Exploratory analysis

6.2 DATA SCIENCE - DEFINITION

Data Science is a multidisciplinary science with the objective of performing data analysis to generate knowledge that can be used for decision-making. This knowledge can be in the form of similar patterns or predictive planning models, forecasting models etc. A data science application collects data and information from multiple heterogeneous sources, cleans, integrates, processes and analyses this data using various tools and presents information and knowledge in various visual forms.

As stated earlier, data science is multidisciplinary, as shown in Figure 1.

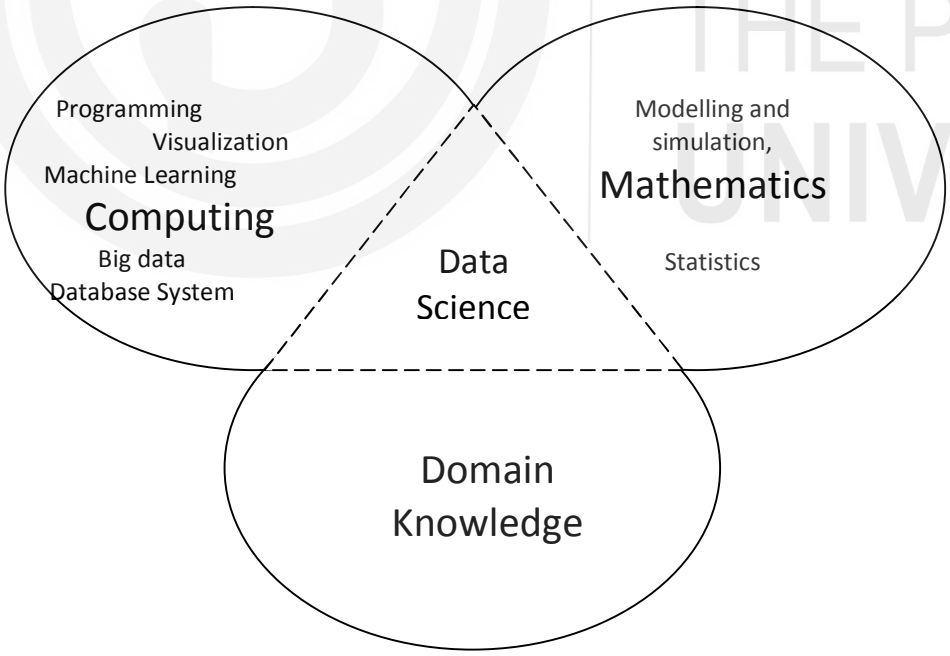

Figure 6.1: Data Science

What are the advantages of Data science in an organisation? The following are some areas in which data science can be useful.

- It helps in making business decisions, such as deciding the health of companies with whom they plan to collaborate,

- It may help make better predictions for the future, such as creating strategic plans for the company based on present trends etc.
- It may identify similarities among various data patterns leading to applications like fraud detection, targeted marketing etc.

Data science is a way forward for business decision-making, especially in the present-day world, where data is generated at the rate of Zetta bytes.

Data Science can be used in many organisations. Some of the possible uses of data science are given below:

- It has excellent potential for finding the best dynamic route from a source to a destination. Such applications may constantly monitor the traffic flow and predict the best route based on collected data.
- It may bring down the logistic costs of an organisation by suggesting the best time and route for transporting foods
- It can minimise marketing expenses by identifying similar group buying patterns and performing selective advertising based on the data obtained.
- It can help in making public health policies, especially in the case of disasters.
- It can be useful in studying the environmental impact of various developmental activities
- It can be very useful in saving resources in smart cities

6.3 TYPES OF DATA

The type of data is one of the essential aspects determining the kind of analysis that can be performed on data. In data science, the following are the different types of data that are required to be processed:

1. Structured Data
2. Semi-Structured Data
3. Unstructured data
4. Data Streams

Structured Data

Since the start of the era of computing, the computer has been used as a data processing device. However, it was not before the 1960s when businesses started using computers to process their data. One of the most popular languages of that era was the **Co**mmon **B**usiness-**O**riented Language (COBOL). COBOL had a data division representing the structure of the data being processed. This was followed by a disruptive seminal design of technology by E.F. Codd. This led to the creation of relational database management systems (RDBMS). RDBMS allows structured storage, retrieval and processing of integrated data of an organisation that can be securely shared among several applications. The RDBMS technology also supported secure transactions, thus, became a major source of data generation. Figure 2 shows the sample structure of data that may be stored in a relational database

system. One of the key characteristics of structured data is that it can be associated with a schema. In addition, each schema element may be related to a specific data type.

Customer (*<u>custID</u>, custName, custPhone, custAddress, cost Category, custPAN, custAadhar*)

Account (*<u>AccountNumber</u>,custIDoffirstaccountholder, AccountType, AccountBalance)*

JointHolders (AccountNumber, custID)

Transaction(transDate, transType, AccountNumber, Amountoftransaction)

Figure 6.2: A sample schema of structured data

Relational data is structured data. Many organisations are collecting a large amount of this structured data as the backend of their applications. The concept of a data warehouse was introduced in the 90s. A data warehouse is a time-invariant, subject-oriented aggregation of data of an organisation that can be used for decision-making. Data in a data warehouse is represented using dimension tables and fact tables. The dimensional tables classify the data of fact tables. The data of the data warehouse is also structured in nature and can be used for analytical data processing and data mining. In addition, many different types of database management systems have been developed, mostly storing structured data.

However, with the growth of communication and mobile technologies, many different applications became very popular, leading to the generation of a very large amount of semi-structured and unstructured data. These are discussed next.

Semi-structured Data

As the name suggests, Semi-structured has some structure in it. The structure of semi-structured data is due to the use of tags or key/value pairs. The common form of semi-structured data is produced through XML, JSON objects, Server logs, EDI data, etc. An example of semi-structured data is shown in Figure 3.

<Book>
<title>Data Science and Big Data</title>
<author>R Raman</author>
<author>C V Shekhar</author>
<yearofpublication>2020</yearofpublication>
</Book>
	"Book": {
	"Title": "Data Science",
	"Price": 5000,
	"Year": 2020
	}

Figure 6.3: Sample semi-structured data

Unstructured Data

The unstructured data does not follow any schema definition. For example, the written text-like content of this Unit is unstructured. You may add specific headings or metadata for unstructured data. In fact, the growth of the Internet has resulted in the generation of Zetta bytes of unstructured data. Some of the unstructured data is listed below:

- A large amount of written textual data such as email data, social media data, etc.
- Unprocessed audio and video data
- Image data and mobile data
- Unprocessed natural speech data
- Unprocessed geographical data

This data generally requires enormous storage space, newer processing methods and faster processing capabilities.

Data Streams

A data stream is characterised by a sequence of data over a period of time. Such data may be structured, semi-structured or unstructured, but it gets generated repeatedly. For example, IoT devices like weather sensors will generate a data stream of pressure, temperature, wind direction, wind speed, humidity etc., for a particular place where it is installed. Such data is huge for many applications that are required to be processed in real-time. In general, not all the data of streams need to be stored, and such data is required to be processed for a specific duration of time.

6.3.1 Statistical Data Types

Two distinct types of data can be used in statistical analysis. These are – Categorical data and Quantitative data.

Categorical or qualitative Data:

Categorical data identifies the category of data. For example, the occupation of a person may take values of the categories "Business", "Salaried". "Others", etc. The categorical data can be of two distinct measurement scales called Nominal and Ordinal, which are given in Figure 4. If the categories are not related, then categorical data is of Nominal data type. For example, the Business category and Salaried categories have no relationship; therefore, it is of Nominal type. However, a categorical variable like age category, defining age in categories "0 or more but less than 26", "26 or more but less than 46", "46 or more but less than 61", and "More than 61" has a specific relationship. For example, a person in the "More than 61" age category is older than anyone in any other age category.

Quantitative Data:

Quantitative data is numeric data that can be used to define different scales of data. Qualitative data is also of two basic types –discrete, representing distinct numbers like 2, 3, 5,…; or continuous, representing continuous values of a given variable. For example, your height can be measured using a continuous scale.

Measurement scale of data

Data are raw facts; for example, student data may include name, Gender, Age, Height of student, etc. The name typically is a distinguishing data that tries to distinctly identify two data items, just like the primary key in a

database. However, the name data or any other identifying data may not be useful for data analysis in data science. The data such as Gender, Age, and Height may be used to answer queries of the kind: Is there a difference in the height of boys and girls in the age range 10-15 years? One of the important questions is how you measure the data so that it is recorded consistently. Stanley Stevens, a psychologist, defined the following four characteristics of any scale that can be measured:

- Every representation of the measure should be unique, referred to as the identity of a value (*IDV*).
- The second characteristic is the magnitude (*M*), which clearly can be used to compare the values; for example, a weight of 70.5 kg is more than 70.2 kg.
- The third characteristic is about equality in intervals (*EI*) used to represent the data; for example, the difference between 25 and 30 is 5 intervals, which is the same as the difference between 41 to 46, which is also 5 intervals.
- The last characteristic is about a defined minimum or zero value (*MZV*); for example, in the Kelvin scale, the temperature has an absolute zero value, whereas the Intelligent quotient cannot be defined as zero.

Based on these characteristics, four basic measurement scales are defined. Figure 4 illustrates these measurements, their features and examples.

Measurement Scale	Characteristics				Example
	IDV	*M*	*EI*	*MZV*	
Nominal	Yes	No	No	No	Gender F - Female M- Male
Ordinal	Yes	For rank ordering	No	No	A hypothetical Income Category: 1 - "0 or more but less than 26" 2 -"26 or more but less than 46" 3 - "46 or more but less than 61" 4 - "More than 61"
Interval	Yes	Yes	Yes	No	IQ, Temperature in Celsius
Ratio	Yes	Yes	Yes	Yes	The temperature in K, Age

Figure 6.4: Measurement Scales of Data

6.3.2 Sampling

In general, the size of the data that is to be processed is quite large. This leads you to the question: whether you would use the entire data or some representative sample of this data?. In several data science techniques, sample data is used to develop an exploratory model. Thus, even in data science, sampling is one of the ways to enhance the speed of exploratory data analysis. In this case, the population may be the entire data set you may be interested in. Figure 5 shows the relationships between the population and the sample. One of the questions that can be asked in this context is, what should

be the size of a good sample? You may have to find the answer in the literature. However, you may please note that a good sample is representative of its population.

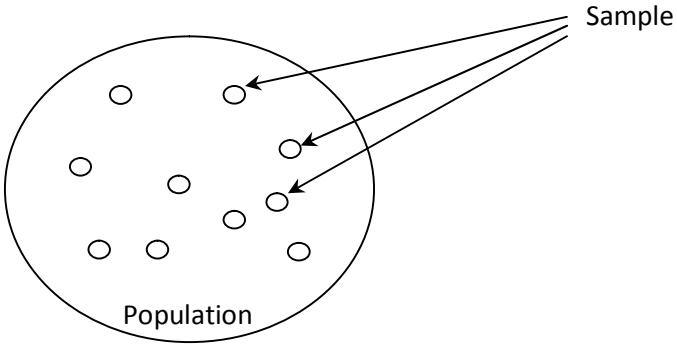

Figure 6.5: Population and Sample

One of the main objectives of statistics, which uses sample data, is to determine the statistics of the sample and find the probability that the statistic developed for the sample would determine the parameters of the population with a specific percentage of accuracy. Please note the terms stated above are very important and explained in the following table:

Term	Used for	Example
Statistic	The statistic is computed for a sample	Sample mean ($\underline{x}$), Sample Standard deviation (s), Sample size (n)
Parameter	Parameters are predicted from a sample and are about the population	Population mean (μ), Population Standard deviation (σ), Population size (N)

Next, we discuss different kinds of analysis that can be performed on data.

Check Your Progress 1:

1. Define the term data science.

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2. Differentiate between structured, semi-structured, unstructured and stream data.

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3. What would be the measurement scale for the following? Give reasons in support of your answer.

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Age, AgeCategory, Colour of the eye, weight of students of a class, Grade of students, 5-point Likert scale

6.4 BASIC METHODS OF DATA ANALYSIS

The data for data science is obtained from several data sources. This data is first cleaned of errors, duplication, aggregated and then presented in a form that can be analysed by various methods. This section defines some of the basic techniques used for analysing data. These are Descriptive analysis, Exploratory data analysis and Inferential data analysis.

6.4.1 Descriptive Analysis

Descriptive analysis is used to present basic data summaries; however, it does not attempt to interpret the data. These summaries may include different statistical values and certain graphs. Different types of data are described using different ways. The following example illustrates this concept:

Example 1: Consider the data given in the following Figure 6. Show the summary of categorical data in this Figure.

Enrolment Number	Gender	Height
S20200001	F	155
S20200002	F	160
S20200003	M	179
S20200004	F	175
S20200005	M	173
S20200006	M	160
S20200007	M	180
S20200008	F	178
S20200009	F	167
S20200010	M	173

Figure 6.6: Sample Height Data

Please note that the variable *enrolment number* need not be used in the analysis, so no summary data for the enrolment number will be performed.

Descriptive Analysis of Categorical Data:

Gender is a categorical variable in Figure 6. In this case, the summary would be the frequency table of various categories. For example, for the given data, the frequency distribution would be:

Gender	Frequency	Proportion	Percentage
Female (F)	5	0.5	50%
Male (M)	5	0.5	50%

In addition, you can draw a bar chart or pie chart to describe the data of the Gender variable. The pie chart for such data is shown in Figure 7. Details of different charts are explained in Unit 4. In general, you draw a bar graph if the number of categories exceeds three.

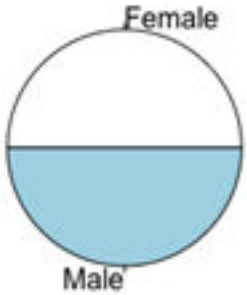


Figure 6.7: The Pie Chart

Descriptive analysis of Quantitative Data:

The height is a quantitative variable. The descriptive analysis of quantitative data is presented by the following:

1. Describing the central tendencies of the data
2. Describing the spread of the data.

Central tendencies of Quantitative data: Mean and Median are two basic measures that define the centre of data by using different ways. They are explained below with the help of an example.

Example 2: Find the mean and median of the following data:

Data Set (*n* observations)	1	2	3	4	5	6	7	8	9	10	11
x	4	21	25	10	18	9	7	14	11	19	14

The mean can be computed as follows:

$$\bar{x} = \frac{\sum_{i=1}^{n} x}{n}$$

For the given data

$$\bar{x} = \frac{(4 + 21 + 25 + 10 + 18 + 9 + 7 + 14 + 11 + 19 + 14)}{11}$$

Mean, $\bar{x} = 13.82$

The median of the data would be the mid-value of the sorted data. First, data is sorted, and the median is computed using the following formula:

If *n* is even, then

median = $[(Value\ of\ (\frac{n}{2})^{th})\ position + Valueof\ ((\frac{n}{2} + 1)^{th} position]/2$

If *n* is odd, then

median =$(Value\ of\ (\frac{n+1}{2})^{th})\ position$

For this example, the sorted data is as follows:

Data Set (n observations)	1	2	3	4	5	6	7	8	9	10	11
x	4	7	9	10	11	14	14	18	19	21	25

So, the median is:

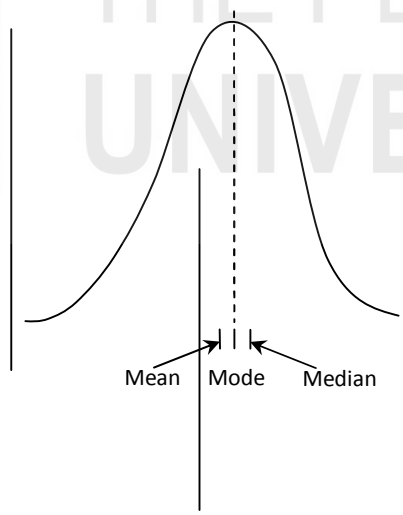
median = $(Value\ of\ (\frac{11+1}{2})^{th})\ position$ = 14

Please note that outliers, defined as values highly different from most other values, can impact the mean but not the median. For example, if one observation in the data, as shown in example 2, is changed as:

Data Set (n observations)	1	2	3	4	5	6	7	8	9	10	11
x	4	7	9	10	11	14	14	18	19	21	100

Then, the median will remain at 14; however, the mean will change to 20.64, which is quite different from the earlier mean. Thus, you should be careful about the presence of outliers during data analysis.

Interestingly, mean and mode may be useful in determining the nature of data. The following table describes these conditions:

Relationship between mean and mode	**Comments about observations**	**A Possible Graph of Data Distribution**
Almost Equal values of mean and median	The distribution of data may be symmetric	

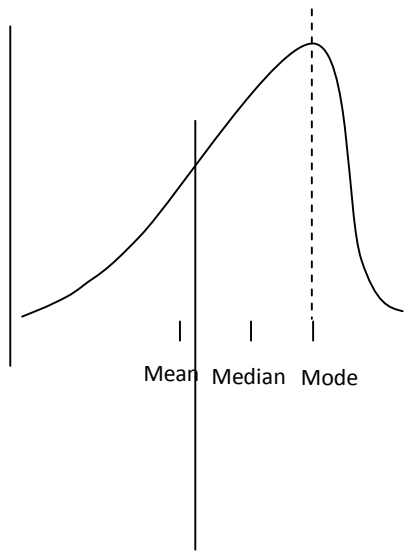
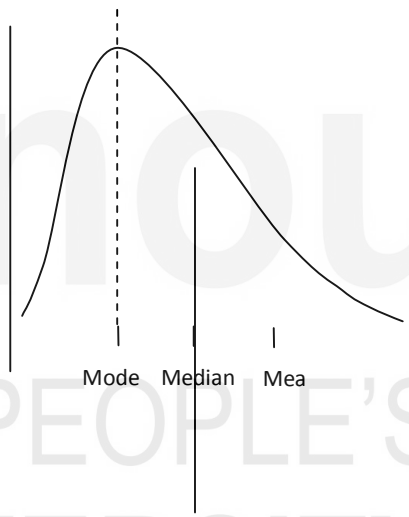
Mean >> Median	The distribution may be left-skewed	
Mean << Median	The distribution may be right-skewed	

Figure 6.8: Mean and Median for possible data

The concept of data distribution is explained in the next Unit.

Mode: Mode is defined as the most frequent value of a set of observations. For example, in the data of example 2, the value 14, which occurs twice, is the mode. The mode value need not be a mid-value; rather, it can be any value of the observations. It just communicates the most frequently occurring value only. In a frequency graph, the mode is represented by the peak of the data. For example, in the charts shown in Figure 8, the value corresponding to the peaks is the mode.

Spread of Quantitative Data: Another important aspect of defining quantitative data is its spread or variability of observed data. Some of the measures for the spread of data are given in Figure 9.

Measure	Description	Example (Please refer to Data of Example 2)
Range	Minimum to Maximum Value	4 to 25
Variance	The sum of the squares of the difference between the observations and its sample mean is divided by (n-1), as the difference of nth value can be determined from (n-1) computed differences since the overall sum of differences has to be zero. The formula of variance for a sample is: $$s^2 = \frac{1}{(n-1)}\sum_{i=1}^{n}(x - \underline{x})^2$$ However, in case you are determining the Population variance, then you can use the following formula: $$\sigma^2 = \frac{1}{N}\sum_{i=1}^{n}(x-\mu)^2$$	Try both the formula and then match the answer: 40.96
Standard Deviation	Standard deviation is one of the most used measures for finding the spread or variability of data. It can be computed as: For Sample: $$s = \sqrt{\frac{1}{(n-1)}\sum_{i=1}^{n}(x - \underline{x})^2}$$ For Population: $$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{n}(x-\mu)^2}$$	Try both the formula and then match the answer: 6.4
5-Point Summary and Interquartile Range (IQR)	To create a 5-point summary, first, you need to sort the data. The five-point summary is defined as follows: Minimum Value (Min) 1st Quartile<=25% values (Q1) 2nd Quartile is median (M) 3rd Quartile is <=75% values (Q3) Maximum Value (Max) IQR is the difference between the values of the 3rd and 1st quartiles.	Use Sorted data from Example 2 Min = 4 Q1=(9+10)/2=9.5 M = 14 Q3=(18+19)/2=18.5 Max=25 IQR= 18.5-9.5=9

Figure 6.9: The Measure of Spread or Variability

The IQR can also be used to identify suspected outliers. In general, a suspected outlier can exist in the following two ranges:

Observation/values less than Q1 – 1.5·IQR

Observation/values more than Q3+1.5 · IQR

For example 2, IQR is 9; therefore, the outliers may be: Values < (9.5 – 9) or Values < 0.5. or Values > (18.5 – 9) or Values > 27.5.

Thus, there is no outlier in the initial data of Example 2.

You may draw various plots for the qualitative data, such as histogram, boxplot, etc. These plots are explained in Unit 4 of this block.

Check Your Progress 2

1. The age category of students is categorical data. What information would you like to show for its descriptive analysis?

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2. Age is quantitative data; how will you describe its data?

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3. How can you find that given data is left skewed?

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4. What is IQR? Can it be used to find outliers?

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6.4.2 Exploratory Analysis

Exploratory data analysis was suggested by John Turkey of Princeton University in 1960 as a group of methods that can be used to learn the possibilities of relationships amongst data. After obtaining relevant data for analysis, instead of performing the final analysis, you may like to explore the data for possible relationships using exploratory data analysis. In general, graphs are some of the best ways to perform exploratory analysis. Some of the standard methods that you can perform during exploratory analysis are as follows:

1. As a first step, you may perform the descriptive analysis of various categorical and qualitative variables of your data. Such information is

very useful in determining the suitability of data for analysis. This may also help you in data cleaning, modification and transformation of data.

a. For the qualitative data, you may create frequency tables and bar charts to know the distribution of data among different categories. A balanced distribution of data among classes is most desirable. However, such distribution may not be possible in actual situations. Several methods have been suggested to deal with such situations. Some of those will be discussed in the later units.

b. For the quantitative data, you may compute the mean, median, standard deviation, skewness and kurtosis. The kurtosis value relates to peaks (determined by mode) in the data. In addition, you may also draw the charts like histogram to look into frequency distribution.

2. Next, after performing the univariate analysis, you may try to perform some bi-variate analysis. Some of the basic statistics that you can perform for bi-variate analysis include the following:

   a. Make a two-way table between categorical variables and make related stacked bar charts. You may also use chi-square testing to find any significant relationships.

   b. You may draw side-by-side box plots to check if the data of various categories have differences.

   c. You may draw a scatterplot and check the correlation coefficient if that exists between two variables.

3. Finally, you may like to look into the possibilities of multi-variate relationships among data. You may use dimensionality reduction by using techniques like feature extraction or principle component analysis, you may perform clustering to identify possible sets of classes in the solution space, or use graphical tools, like bubble charts, to visualise the data.

It may be noted that exploratory data analysis helps in identifying the possibilities of relationships amongst data but does not promise that a causal relationship may exist amongst variables. The causal relationship has to be ascertained through qualitative analysis. Let us explain the exploratory data analysis with the help of an example.

Example 3: Consider the sample data of students given in Figure 6 about Gender and Height. Let us explore this data for an analytical question: Does height depend on Gender?

You can perform the exploratory analysis on this data by drawing a side-by-side boxplot for Male and Female students' height. This boxplot is shown in Figure 6.10.

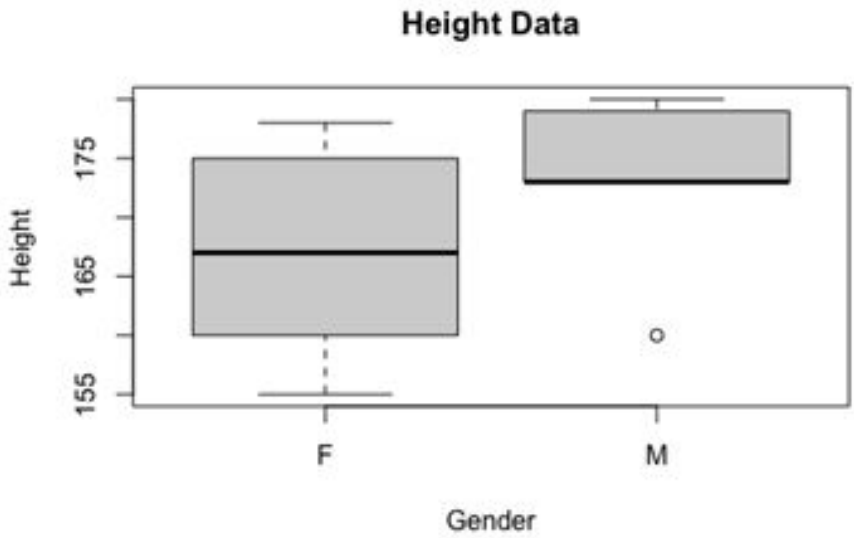

Figure 6.10: Exploratory Data Analysis

Please note that the box plot of Figure 10 shows that the average height of male students is more than that of female students. Does this result apply, in general, to the population? To answer this question, you need to find the probability of the occurrence of such sample data, thereafter, you may perform an Inferential analysis.

Check Your Progress 3

1. What is the advantage of using a boxplot?

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6.5 SUMMARY

This Unit introduces basic statistical and analytical concepts of data science. This Unit first introduces you to the definition of data science. Data science uses concepts from computing, mathematics and domain knowledge. The types of data for data science are defined in two different ways. First, it is defined based on the structure and generation rate of data. Next, it is defined as the measures that can be used to capture the data. In addition, the concept of sampling has been defined in this Unit.

This Unit also explains some of the basic analysis techniques, including descriptive, exploratory, inferential and predictive. A few misconceptions related to data science have also been explained with the help of examples. This Unit also introduces you to some of the applications of data science and the data science life cycle. In the ever-advancing technology, it is suggested to keep reading about newer data science applications.

6.6 SOLUTIONS/ANSWERS

Check Your Progress 1:

1. Data science integrates the principles of computer science, mathematics and domain knowledge to create mathematical models that show relationships among data attributes. In addition, data science uses data to perform predictive analysis.

2. Structured data has a defined dimensional structure identified by attributes, for example, tables, data cubes, etc. Semi-structured data has some structure due to the use of tags; however, this structure may be flexible, for example, XML data. Unstructured data has no structure at all, like long texts. Data streams, on the other hand, may be structured, semi-structured or unstructured data being produced continuously.

3. Age category would be categorical data, it will be of the ordinal scale, as there are differences among categories, but that difference cannot be defined quantitatively. The weight of the students in a class is a ratio scale. The Grade is also a measure of the ordinal scale. The 5-point Likert scale is also ordinal data.

Check Your Progress 2:

1. Descriptive analysis of categorical data may include the total number of observations, frequency table and bar or pie chart.

2. The descriptive age may include mean, median, mode, skewness, kurtosis, standard deviation and histogram or box plot.

3. For left-skewed data, the mean is substantially higher than the median and mode.

4. The difference between the Quartile 3 and Quartile 1 is the interquartile range (IQR). In general, suspected outliers are at a distance of 1.5 times IQR higher than the 3rd quartile or 1.5 times IQR lower than the 1st quartile.

Check Your Progress 3:

1. Boxplot shows a 5-point summary of data. A well-spread box plot is an indicator of normally distributed data. Side-by-side boxplots can be used to make a comparison of scale data values of two or more categories.

UNIT 7 INFERENTIAL AND PREDICTIVE DATA ANALYSIS – AN INTRODUCTION

Structure

7.0 Introduction

7.1 Objectives

7.2 Inferential Analysis

7.3 Predictive Analysis

7.4 Common Misconceptions of Data Analysis

7.5 Summary

7.6 Solutions/Answers

7.0 INTRODUCTION

In today's data-driven world, the ability to extract meaningful insights from data is essential across various domains. Data analysis goes beyond simply organizing and summarizing data—it involves drawing conclusions, making predictions, and avoiding pitfalls in interpretation. This unit focuses on three key aspects of advanced data analysis: **Inferential Data Analysis**, **Predictive Data Analysis**, and the **Common Misconceptions of Data Analysis**.

Inferential data analysis enables us to make judgments about a population based on sample data. It includes tools and techniques such as hypothesis testing, confidence intervals, and significance testing, allowing analysts to move from observed data to broader generalizations with a quantifiable level of certainty.

Predictive data analysis, on the other hand, leverages historical data to forecast future outcomes. This approach uses statistical models and machine learning techniques to identify patterns and trends that can inform decision-making and strategic planning.

Despite the power of these analytical methods, there are common misconceptions that can lead to erroneous conclusions. Misunderstanding the role of correlation, misinterpreting p-values, and overreliance on models without validation are just a few examples. This unit aims to clarify these misconceptions, promoting a more accurate and responsible approach to data analysis.

Following the structured progression from data preparation in Unit 5 to descriptive and exploratory analysis in Unit 6, Unit 7 takes the learner deeper into the data science pipeline by introducing **inferential and predictive data analysis**. While Unit 5 ensured the data was clean, integrated, and ready for use, and Unit 6 enabled learners to summarize and explore its structure and patterns, Unit 7 builds upon these outcomes to help learners make statistically sound inferences and forward-looking predictions.

Inferential analysis allows students to draw conclusions about populations based on sample data, using tools such as hypothesis testing and estimation. These skills are fundamental in validating assumptions and confirming whether observed patterns are statistically significant or due to random chance. Predictive analysis, on the other hand, focuses on using historical data to model and forecast future outcomes—a vital skill in industries like finance, marketing, healthcare, and supply chain management.

This unit also helps learners develop critical thinking by addressing common misconceptions in data interpretation and analysis, thus promoting a more rigorous and evidence-based approach to decision-making. With a strong foundation from Unit 5 and Unit 6 learners are well-positioned to carry out inferential and predictive tasks with accuracy and confidence.

By the end of this Unit 7, students will be able to move beyond descriptive summaries and begin generating actionable insights and predictions. This marks a significant step toward real-world problem-solving and reflects the increasing demand for professionals who can convert raw data into future-facing strategies through statistical and analytical expertise.

7.1 OBJECTIVES

At the end of this Unit, you should be able to:

- Understand and describe Inferential Analysis
- Understand and describe Predictive Analysis
- Understand and describe the common mistakes in data analysis

7.2 INFERENTIAL ANALYSIS

Inferential Data Analysis refers to the process of drawing conclusions about a population based on data collected from a sample. It uses probability theory to estimate population parameters, test hypotheses, and make predictions or generalizations. Unlike descriptive statistics, which summarize data, inferential statistics allow us to make decisions or inferences about populations based on limited sample data. This section explores the foundational concepts and techniques involved in inferential analysis, with suitable numerical examples to aid understanding. Some of the foundational concepts are as follows:

1. Population and Sample

A fundamental idea in inferential analysis is the distinction between a population and a sample. A population is the complete set of individuals, items, or data under consideration, while a sample is a subset of the population selected for analysis. For example, suppose a university has 10,000 students. To estimate the average GPA of all students, we might randomly select a sample of 100 students. The insights gained from this sample can then be generalized to the entire population using inferential techniques.

2. Estimation

Estimation involves inferring the value of a population parameter based on sample data. There are two types of estimation: point estimation and interval estimation.

In point estimation, a single value is used as the estimate of a population parameter. For instance, if the average GPA of a sample of 100 students is 3.2, then 3.2 is a point estimate of the population mean GPA.

Interval estimation, on the other hand, provides a range within which the true population parameter is likely to fall, along with a confidence level. For example, if the sample mean GPA is 3.2 with a standard deviation of 0.4 and a sample size of 100, we can construct a 95% confidence interval using the formula:

$CI = \bar{x} \pm z * (\sigma / \sqrt{n})$

Substituting the values gives:

$CI = 3.2 \pm 1.96 * (0.4 / \sqrt{100}) = 3.2 \pm 0.0784$

So, the confidence interval is (3.1216, 3.2784). This means we are 95% confident that the true population mean lies within this interval.

3. Hypothesis Testing

Hypothesis testing is a structured process used to test assumptions or claims about population parameters. It involves several steps:

a. State the null hypothesis (H_0) and the alternative hypothesis (H_1).
b. Choose a significance level (α), typically 0.05.
c. Compute the test statistic based on sample data.
d. Compare the test statistic to a critical value or calculate the p-value.
e. Make a decision to reject or fail to reject the null hypothesis.

For example, suppose we wish to test the claim that the average GPA of students is 3.0. We take a sample and find a mean of 3.2, a standard deviation of 0.4, and a sample size of 100. The null hypothesis is H_0: $\mu = 3.0$, and the alternative is H_1: $\mu \neq 3.0$. The z-statistic is calculated as:

$z = (\bar{x} - \mu) / (\sigma / \sqrt{n}) = (3.2 - 3.0) / (0.4 / \sqrt{100}) = 0.2 / 0.04 = 5.0$

The critical z-value for a 5% significance level is ±1.96. Since 5.0 exceeds this threshold, we reject the null hypothesis and conclude that the average GPA is significantly different from 3.0.

4. Confidence Intervals (Revisited)

Confidence intervals are used not only for means as discussed above in Estimation, but also for proportions. Suppose in a survey, 60 out of 100 people say they prefer online learning. The sample proportion is $\hat{p} = 0.6$. A 95% confidence interval for the population proportion can be calculated using:

$CI = \hat{p} \pm z * \sqrt{[\hat{p}(1 - \hat{p}) / n]} = 0.6 \pm 1.96 * \sqrt{(0.6 * 0.4 / 100)} = 0.6 \pm 0.096$

Thus, the confidence interval is (0.504, 0.696), indicating that the true proportion likely falls within this range.

5. p-value and Significance

The p-value represents the probability of observing a test statistic as extreme as, or more extreme than, the one observed, assuming the null hypothesis is true. If the p-value is less than the chosen significance level (α), we reject the null hypothesis.

Returning to our earlier GPA example for Hypothesis testing where the z-value was 5.0, the corresponding p-value is less than 0.0001. Since this is below 0.05, we reject H_0, supporting the claim that the mean GPA is not 3.0.

6. t-test and z-test

Both t-tests and z-tests are used to compare sample means, but their applicability depends on certain conditions. The z-test is used when the population standard deviation is known and the sample size is large (typically $n > 30$). The t-test is used when the population standard deviation is unknown and/or the sample size is small.

For example, suppose we have a sample of 25 students with a sample mean score of 75 and a sample standard deviation of 10. We wish to test whether the mean score differs from 70. The t-statistic is:

$t = (75 - 70) / (10 / \sqrt{25}) = 5 / 2 = 2.5$

With 24 degrees of freedom, the critical t-value at a 5% level is approximately 2.064. Since $2.5 > 2.064$, we reject the null hypothesis and conclude that the mean score is significantly different from 70.

7. Chi-Square Test

The chi-square test is used to assess relationships between categorical variables or to test the goodness of fit. For instance, suppose we survey 100 students on their preferred learning mode and obtain the following results: 50 prefer online, 30 prefer offline, and 20 prefer hybrid. If we expect equal preference, the expected count for each category is 33.33. The chi-square statistic is computed as:

$\chi^2 = \Sigma[(O - E)^2 / E] \approx (50 - 33.33)^2 / 33.33 + (30 - 33.33)^2 / 33.33 + (20 - 33.33)^2 / 33.33 \approx 14.16$

With 2 degrees of freedom, the critical value at $\alpha = 0.05$ is 5.99. Since $14.16 > 5.99$, we reject the null hypothesis and conclude that student preferences are not equally distributed.

8. ANOVA (Analysis of Variance)

ANOVA is used when comparing the means of three or more groups to determine if at least one group mean is significantly different. Consider a study comparing test scores of students taught using three different methods:

Method A: [85, 90, 88]

Method B: [78, 75, 80]

Method C: [92, 88, 91]

We compute the between-group variance and within-group variance, calculate the F-ratio, and compare it to the critical F-value from the F-distribution table. If the F-ratio exceeds the critical value, we reject the null hypothesis and conclude that not all teaching methods yield the same average score.

From the above discussion it can be concluded that Inferential Data Analysis equips us with the tools to make informed decisions and generalizations based on sample data. Concepts such as hypothesis testing, confidence intervals, and various statistical tests (z-test, t-test, chi-square, and ANOVA) allow researchers and analysts to assess claims, identify trends, and support conclusions with statistical evidence. A solid understanding of these techniques is essential for responsible and effective data analysis.

In short, the inferential analysis is performed to answer the question - what is the probability that the results obtained from an analysis can be applied to the entire population?

You can perform many different types of statistical tests on data. Some of the well-known tools for data analysis are listed in Table-1 below.

Table 7.1: Some tools for data analysis

Test	Why Performed?
Univariate Analysis: Z-Test or T-test	To determine if the computed value of the mean of a sample can be applied to the population and related confidence interval.
Bivariate Chi-square test	To test the relationship between two categorical variables or groups of data
Two sample T-Test	To test the difference between the means of two variables or groups of data
One way ANOVA	To test the difference in mean of more than two variables or groups of data
F-Test	It can be used to determine the equality of variance of two or more groups of data
Correlation analysis	Determines the strength of the relationship between two variables
Regression analysis	Examines the dependence of one variable over a set of independent variables
Decision Trees	Supervised learning used for classification
Clustering	Non-supervised Learning

Some of the tests mentioned in Table-1, like the Regression analysis, decision trees etc. are related to predictive analysis, which is discussed in the subsequent section

Check Your Progress 1:

1. What is the main objective of inferential data analysis?

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2. How does a confidence interval help in inference?

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3. When is a t-test preferred over a z-test?

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4. What does a p-value indicate in hypothesis testing?

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5. What does the chi-square test assess?

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7.3 PREDICTIVE ANALYSIS

Predictive Data Analysis involves predictions about future or unknown events using historical data to make. It combines statistical techniques, data mining, and machine learning algorithms to identify patterns and relationships in data. These patterns are then used to build models that can forecast outcomes with a certain level of confidence. Predictive analytics is widely used in domains such as finance, marketing, healthcare, and operations to support decision-making and strategic planning.

1. **Regression Analysis:** Regression analysis is a fundamental statistical method used in predictive data analysis to model and analyze the relationships between a dependent variable and one or more independent variables. Its primary goal is to predict future values based on historical data patterns.

   In predictive analysis, regression helps estimate the expected value of a target variable (such as sales, demand, or performance) by identifying how changes in input variables (like advertising spend, temperature, or

hours studied) influence it. The most commonly used form is linear regression, where the relationship is modelled as a straight line. For more complex situations, multiple regression is used to account for several predictors simultaneously.

For example, a company can use regression analysis to forecast monthly sales based on factors such as marketing expenses and number of sales calls. The resulting model can provide actionable insights and guide data-driven decision-making.

Regression models are valued for their interpretability and simplicity, making them a starting point in many predictive analytics workflows.

a) Linear Regression : This method models the relationship between two variables by fitting a linear equation.

Example: A company wants to predict sales based on advertising expenditure.
Advertising (X): 10, 15, 20, 25
Sales (Y): 25, 35, 45, 55
Using linear regression, the best-fit line: $Y = a + bX$
Slope $b \approx 2$, Intercept $a \approx 5 \rightarrow Y = 5 + 2X$
To predict sales for advertising budget 30: $Y = 5 + 2(30) = 65$ units

In this example, a company wants to predict its sales based on the amount it spends on advertising. The historical data shows advertising expenditures of 10, 15, 20, and 25 thousand rupees corresponding to sales of 25, 35, 45, and 55 units respectively. Linear regression is used to model the relationship between advertising expenditure (independent variable) and sales (dependent variable) with the equation $Y = a + bX$, where Y represents sales, X represents advertising spend, a is the intercept, and b is the slope. By analyzing the data, the slope is found to be approximately 2, indicating that for every additional thousand rupees spent on advertising, sales increase by 2 units. The intercept is approximately 5, meaning that if no money is spent on advertising, the expected sales would be about 5 units. Therefore, the regression equation becomes $Y = 5 + 2X$. Using this equation, if the company plans to spend 30 thousand rupees on advertising, the predicted sales would be $Y = 5 + 2(30) = 65$ units. This linear model effectively captures the positive trend between advertising expenditure and sales, enabling the company to make data-driven predictions about future sales based on planned advertising budgets.

b) Multiple Linear Regression

Used when there are multiple predictors.

Example: Predict house price based on size and number of bedrooms.
Size (X_1): 1500, 1800, 2400, 3000; Bedrooms (X_2): 3, 4, 4, 5
Price (Y): 300,000; 350,000; 400,000; 500,000
Model: $Y = a + b_1X_1 + b_2X_2$

Assuming: $Y = 50{,}000 + 100X_1 + 10{,}000X_2$
For a 2000 sq ft house with 4 bedrooms: $Y = 290{,}000$

In this example, the goal is to predict the price of a house based on two factors: its size in square feet (X_1) and the number of bedrooms (X_2). The historical data includes house sizes of 1500, 1800, 2400, and 3000 square feet, with corresponding bedroom counts of 3, 4, 4, and 5. The house prices for these examples are 300,000; 350,000; 400,000; and 500,000 units respectively. Multiple linear regression is used to model the relationship, with the formula $Y = a + b_1X_1 + b_2X_2$, where Y represents the house price, a is the intercept, b_1 is the coefficient for house size, and b_2 is the coefficient for the number of bedrooms. Assuming the regression equation is $Y = 50{,}000 + 100X_1 + 10{,}000X_2$, the intercept 50,000 represents a base price, the coefficient 100 means that for every additional square foot, the price increases by 100 units, and the coefficient 10,000 means that each additional bedroom adds 10,000 units to the price. To predict the price of a house that is 2000 square feet with 4 bedrooms, substitute the values into the equation: $Y = 50{,}000 + 100(2000) + 10{,}000(4)$, which calculates to $Y = 50{,}000 + 200{,}000 + 40{,}000 = 290{,}000$ units. This predicted value gives an estimate of the house price based on its size and number of bedrooms using the linear model.

2. **Classification Models:** Classification models are a fundamental type of supervised machine learning used to predict categorical outcomes. These models are trained on datasets that include both input features and known output labels, enabling them to learn patterns and relationships within the data. The goal is to use this learned information to predict the class or category of new, unseen data points. Classification can be binary, where there are only two possible classes (such as pass/fail or yes/no), or multiclass, involving more than two possible outcomes (like classifying an email as primary, social, or promotional). Common algorithms used in classification tasks include logistic regression, which models the probability of a binary outcome using a logistic function; decision trees, which split data based on feature values to reach a decision; and random forests, which are ensembles of decision trees that enhance accuracy. Other methods include support vector machines that find optimal boundaries between classes, k-nearest neighbours that classify data based on proximity to labeled examples, naive Bayes which applies Bayes' theorem assuming feature independence, and neural networks, which are especially useful for complex tasks such as image and speech recognition.

The performance of classification models is typically evaluated using metrics such as accuracy, which measures the proportion of correct predictions, and precision and recall, which are especially important when dealing with imbalanced datasets. The F1 score, a harmonic mean of precision and recall, provides a balanced measure of model performance. A confusion matrix can also be used to visualize the number of correct and incorrect predictions across each class. Classification models have a broad range of applications across

industries. They are commonly used for tasks like spam detection in emails, diagnosing diseases in healthcare, assessing credit risk in finance, recognizing images and speech, and detecting fraud. By enabling automated decision-making based on historical data, classification models play a critical role in many modern technological systems.

a) Logistic Regression: Used for binary outcomes.
Example: Predict pass (1) or fail (0) based on study hours.

Hours (X): 2, 4, 6, 8, 10;

Pass/Fail (Y): 0, 0, 1, 1, 1
Model: P(pass) = 1 / (1 + e^-(a + bX))
Suppose: P(pass) = 1 / (1 + e^−(−4 + 0.8X))
For X=5: P(pass) = 1 / (1 + e^0) = 0.5

This example demonstrates how logistic regression can be used to predict a binary outcome—in this case, whether a student will pass (represented by 1) or fail (represented by 0) an exam based on the number of hours they studied.

The data is as follows:
- Study Hours (X): 2, 4, 6, 8, 10
- Pass/Fail (Y): 0, 0, 1, 1, 1

This data suggests that as the number of study hours increases, the likelihood of passing the exam also increases.

In logistic regression, the probability of the outcome (pass) is modelled using the logistic (sigmoid) function:

P(pass) = 1 / (1 + e^-(a + bX))

Here, 'a' and 'b' are parameters (also known as the intercept and slope, respectively) that are learned from the data. Suppose we have already determined these parameters, and the model is given by:

P(pass) = 1 / (1 + e^−(−4 + 0.8X))

To make a prediction for a student who studied for 5 hours (X = 5), we substitute this value into the model:

P(pass) = 1 / (1 + e^−(−4 + 0.8×5))
= 1 / (1 + e^−(−4 + 4))
= 1 / (1 + e^0)
= 1 / (1 + 1)
= 0.5

This result means that a student who studied for 5 hours has a 50% chance of passing the exam, according to the logistic regression model.

3. **Time Series Forecasting:** Time series forecasting is a method used to predict future values based on previously observed data points collected over time. This technique is especially useful in scenarios where data is recorded sequentially and patterns such as trends, seasonality, or cycles are present. Time series data is common in various domains, including

finance (stock prices), economics (GDP growth), weather prediction, and business analytics (sales forecasting). By analyzing the structure and characteristics of the data, forecasters can build models that anticipate future developments with a reasonable degree of accuracy.

Several techniques are used for time series forecasting, ranging from simple methods like moving averages and exponential smoothing to more advanced statistical and machine learning models such as ARIMA (Auto Regressive Integrated Moving Average), SARIMA (Seasonal ARIMA), and Long Short-Term Memory (LSTM) networks. The choice of method depends on the nature of the data and the complexity of the patterns it exhibits. Accurate time series forecasting helps organizations in decision-making, resource planning, and risk management by offering insights into potential future outcomes based on historical trends.

a) Moving Average: The moving average is a simple forecasting technique used to predict future values by averaging a fixed number of past observations. It smooths out short-term fluctuations and highlights longer-term trends or cycles in the data. This method is particularly useful in time series data such as sales, stock prices, or weather measurements. It *provides a*verages past n values to predict next.

Example: Sales for 5 months: [100, 120, 130, 110, 140] 3-month moving average for month 6: (130 + 110 + 140) / 3 = 126.67

For instance, consider the monthly sales figures for a business over five months: [100, 120, 130, 110, 140]. To predict the sales for the sixth month using a 3-month moving average, we take the average of the most recent three months. That is, we calculate (130 + 110 + 140) / 3, which equals 126.67. This result represents the forecasted sales value for the sixth month, based on the trend observed in the previous three months.

b) Exponential Smoothing

Exponential smoothing is a forecasting technique used to predict future values by giving more weight to recent observations while not entirely discarding older data. This method is particularly effective for data with no clear trend or seasonal pattern. It uses a smoothing constant, denoted by α (alpha), which determines how much weight is given to the most recent observation. The formula for exponential smoothing is: $F_{t+1} = \alpha A_t + (1 - \alpha)F_t$, where F_{t+1} is the forecast for the next period, A_t is the actual value in the current period, F_t is the forecast for the current period, and α is a value between 0 and 1.

Exponential smoothing Gives more weight to recent observations.
Formula: $F_{t+1} = \alpha A_t + (1 - \alpha) * F_t$
Example: $\alpha = 0.5$, $F_1 = 100$, $A_1 = 110 \rightarrow F_2 = 105$

In the above example, suppose the smoothing constant α is 0.5, the forecast for the first period (F_1) is 100, and the actual observed value (A_1)

is 110. Using the formula, the forecast for the second period (F_2) would be calculated as: $F_2 = 0.5 * 110 + (1 - 0.5) * 100 = 55 + 50 = 105$. This result shows that the new forecast is adjusted towards the actual value observed, balancing past expectations with new information.

4. **Decision Trees:** Decision trees are a type of machine learning model used for both classification and regression tasks. They represent decisions and their possible consequences in a tree-like structure, where each internal node represents a decision based on a feature, each branch represents the outcome of that decision, and each leaf node represents a final prediction or classification. Decision trees are intuitive and easy to interpret, making them popular in areas where transparency is important. Decision tree, Models decisions and consequences in tree-like structure.

   Example: Loan approval based on income and credit score. If income > 50,000 and credit score > 700 → Approve; Else → Deny use case: income = 60,000 and score = 720 → Approve

   In the above example, consider a simple decision tree model used to determine whether a loan should be approved based on an applicant's income and credit score. The model might follow a rule such as: if the applicant's income is greater than 50,000 and their credit score is above 700, the loan is approved; otherwise, it is denied. Using this model, if an applicant has an income of 60,000 and a credit score of 720, the decision tree would lead to an approval. This straightforward structure helps organizations make consistent, data-driven decisions.

5. **Model Evaluation Metrics**

   Model evaluation metrics are essential tools in machine learning used to assess how well a model performs on a given dataset. These metrics help determine the accuracy and reliability of predictions made by a model, guiding improvements and comparisons between different models. The choice of metric depends on the type of problem—classification, regression, or clustering. For classification problems, common metrics include accuracy, which measures the proportion of correct predictions; precision, which indicates how many of the predicted positives are truly positive; and recall, which shows how many actual positives were correctly identified. The F1 score, a harmonic mean of precision and recall, is especially useful when dealing with imbalanced datasets. The confusion matrix provides a detailed breakdown of prediction outcomes by showing true positives, false positives, true negatives, and false negatives.

   For regression tasks, metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) are commonly used to quantify the average magnitude of errors between predicted and actual values. R-squared (R^2) is another important metric that indicates the proportion of variance in the dependent variable that is predictable from the independent variables. Evaluating models using the right metrics ensures not only that they are accurate but also that they are suitable for their intended purpose and context.

a) Mean Absolute Error (MAE)

Mean Absolute Error (MAE) is a widely used metric in regression analysis that measures the average absolute difference between predicted values and actual values. Unlike other error metrics that may square or weigh errors, MAE treats all errors equally by taking their absolute values. This makes it a straightforward and interpretable indicator of model performance, showing how far off predictions are from actual outcomes on average.

The formula for MAE is: $MAE = (1/n) \times \Sigma\, |Predicted_i - Actual_i|$

where n is the number of data points, $Predicted_i$ is the predicted value for the i-th data point, and $Actual_i$ is the actual value.

Example: Predicted = [100, 120, 140], Actual = [110, 130, 150] → MAE = 10

In above example, suppose we have three predicted values: [100, 120, 140], and the corresponding actual values are [110, 130, 150]. To calculate MAE, we first find the absolute errors: |100 − 110| = 10, |120 − 130| = 10, and |140 − 150| = 10. Summing these gives a total error of 30. Dividing by the number of observations (3), the MAE becomes 30 / 3 = 10. This result tells us that, on average, the predictions differ from the actual values by 10 units.

b) Root Mean Square Error (RMSE)

The Root Mean Square Error (RMSE) is a standard way to measure the accuracy of a predictive model by calculating the square root of the average of the squared differences between predicted values and actual values.

For example, if we have three prediction errors—each being 10—then the RMSE is calculated by first squaring each error (10^2, 10^2, and 10^2), resulting in 100, 100, and 100. These squared errors are then averaged: (100 + 100 + 100) / 3 = 100. Finally, the square root of this average is taken: $\sqrt{100} = 10$. Therefore, the RMSE in this case is 10. This metric provides a single number that summarizes the magnitude of the prediction errors.

$RMSE = \sqrt{[(10^2 + 10^2 + 10^2) / 3]} = 10$

b) Accuracy (For Classification)

Accuracy is a common metric used to evaluate the performance of classification models. It is calculated as the proportion of correct predictions made by the model out of the total number of predictions. For example, if a model correctly classifies 90 instances out of 100 total instances, the accuracy is computed by dividing the number of correct predictions (90) by the total number of predictions (100), which gives 0.9. When expressed as a percentage, this results in an accuracy of 90%. This means the model correctly predicted the class of the input data 90% of the time.

Correct predictions = 90 out of 100 → Accuracy = 90%

So, we learned that Predictive Data Analysis is a powerful approach to forecast future outcomes using historical data. Techniques such as regression, classification, time series forecasting, and decision trees allow analysts to make data-driven predictions. By evaluating model performance using metrics like MAE, RMSE, and accuracy, we can ensure the reliability of predictive models. A strong grasp of these methods enables informed decision-making in both business and research settings.

Check Your Progress 2:

1. What is the purpose of regression analysis in predictive data analysis?

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2. What is the role of logistic regression?

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3. How is time series forecasting different from other predictive methods?

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4. What does the Mean Absolute Error (MAE) measure?

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5. What is a decision tree in predictive analytics?

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7.4 COMMON MISCONCEPTIONS OF DATA ANALYSIS

Understanding data correctly is essential for making sound decisions in business, science, policy, and everyday life. However, several widespread misconceptions can lead to inaccurate conclusions or misleading results. Three of the most common misconceptions in data analysis are the confusion between correlation and causation, the misinterpretation of trends due to Simpson's Paradox, and the misuse of statistical testing through data dredging.

In this section, we discuss three misconceptions that can affect the result of data science. These misconceptions are explained with the help of an example only.

1. Correlation is Not Causation

One of the most prevalent misconceptions in data analysis is the belief that correlation implies causation. Correlation simply indicates that two variables appear to move together in some predictable pattern—when one changes, the other tends to change as well. However, this observed relationship does not necessarily mean that one variable causes the change in the other. There may be a third variable influencing both, or the correlation could be entirely coincidental.

For example, consider the correlation between ice cream sales and drowning incidents. Data may show that as ice cream sales increase, so do drowning cases. It would be incorrect to conclude that eating ice cream causes drowning. In reality, a confounding variable—hot weather—is responsible for both. During warmer months, people buy more ice cream and also swim more often, which increases the risk of drowning. Thus, while the two variables are correlated, there is no direct causal relationship between them.

This misconception is particularly dangerous in fields such as public health, education, and economics, where assuming causation without rigorous analysis can lead to flawed policies and interventions. Establishing causality requires additional evidence, often through controlled experiments or longitudinal studies.

2. Simpson's Paradox

Simpson's Paradox occurs when a trend that appears in separate groups of data reverses or disappears when those groups are combined. This paradox underscores the importance of considering how data is grouped and the influence of confounding variables. The misleading nature of aggregated data can hide the true story behind subgroup behavior.

An illustrative example of Simpson's Paradox involves university admissions. Suppose overall data shows that men have a higher acceptance rate than women. On the surface, this may suggest gender bias. However, when the data is disaggregated by department, it may reveal that women actually had higher acceptance rates in most individual departments. The apparent discrepancy arises because women tended to apply to more competitive departments with lower acceptance rates, while men applied to less competitive ones. When these departmental differences are ignored, the combined data tells a misleading story.

Simpson's Paradox teaches us to dig deeper and analyze subgroups before drawing conclusions. It reminds analysts to be cautious with aggregated statistics, especially when group characteristics or behavior might differ significantly.

3. Data Dredging (P-Hacking)

Data dredging, often referred to as p-hacking, is the practice of excessively mining data to find statistically significant results without having a prior hypothesis. The danger lies in performing numerous statistical tests and highlighting only those results that appear significant by chance. The more tests performed, the higher the probability of encountering spurious correlations that do not reflect real-world phenomena.

For example, a researcher examining a dataset of lifestyle habits might test 100 different behaviors for their impact on memory. If one test shows that people who eat pickles have significantly better memory, it might seem like a noteworthy finding. However, without a clear hypothesis or statistical correction for multiple comparisons, this result is likely due to chance. It may not replicate in further studies and could mislead both researchers and the public.

The key to avoiding data dredging is disciplined statistical practice: analysts should predefine their hypotheses, limit the number of tests, and apply corrections such as the Bonferroni adjustment when multiple comparisons are made. Findings should also be validated on new or independent datasets to confirm their reliability.

Each of these misconceptions—confusing correlation with causation, misreading aggregated data as shown in Simpson's Paradox, and uncovering misleading patterns through data dredging—highlights the need for critical thinking and methodological rigor in data analysis. Misinterpretation can lead to flawed decision-making, wasted resources, or even harm. By understanding these pitfalls, analysts and data consumers alike can develop a more accurate and nuanced view of what data can truly tell us.

Check Your Progress 3:

1. Why is "correlation is not causation" an important warning in data analysis?

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2. What is Simpson's Paradox?

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3. What is data dredging (p-hacking)?

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4. Give an example of a misleading correlation.

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5. How can analysts avoid falling into the trap of data dredging?

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7.5 SUMMARY

This Unit introduces foundational concepts in data science, focusing on inferential and predictive data analysis. Inferential analysis allows analysts to make generalizations about a population from a sample using statistical techniques such as estimation, hypothesis testing, confidence intervals, t-tests, z-tests, chi-square tests, and ANOVA. It emphasizes the importance of understanding probability to draw conclusions, with examples illustrating how statistical evidence supports or refutes assumptions. Predictive analysis, on the other hand, employs historical data to forecast future outcomes using tools like regression (linear and multiple), classification (logistic regression, decision trees), and time series forecasting (moving averages, exponential smoothing). The section also highlights model evaluation metrics including MAE, RMSE, and accuracy to assess predictive model performance.

Additionally, the unit addresses three common misconceptions in data analysis: mistaking correlation for causation, misinterpretations caused by Simpson's Paradox, and the pitfalls of data dredging (p-hacking). These are explained with illustrative examples to caution against drawing misleading conclusions. The unit underscores the need for rigorous methodology, critical thinking, and validation in data interpretation. Overall, it emphasizes that while data science offers powerful tools for decision-making and prediction, their effectiveness hinges on correct application and awareness of potential analytical errors.

7.6 SOLUTIONS/ANSWERS

Check Your Progress 1:

1. What is the main objective of inferential data analysis?

   Answer: To make generalizations or predictions about a population based on sample data using statistical methods.

2. How does a confidence interval help in inference?

   Answer: It provides a range within which the true population parameter is likely to lie, along with a confidence level, usually 95%.

3. When is a t-test preferred over a z-test?

   Answer: A t-test is used when the population standard deviation is unknown or the sample size is small (typically $n < 30$).

4. What does a p-value indicate in hypothesis testing?

   Answer: It indicates the probability of observing the test statistic as extreme as the one obtained, assuming the null hypothesis is true. A p-value less than 0.05 typically leads to rejecting the null hypothesis.

5. What does the chi-square test assess?

   Answer: It evaluates the relationship between categorical variables or tests how well observed distributions fit expected ones.

Check Your Progress 2:

1. What is the purpose of regression analysis in predictive data analysis? **Answer:** To model the relationship between dependent and independent variables and predict future values.

2. What is the role of logistic regression?

   Answer: Logistic regression predicts binary outcomes, such as yes/no or pass/fail, based on input variables.

3. How is time series forecasting different from other predictive methods? **Answer:**It uses sequential, time-ordered data to predict future values, capturing trends and seasonality.

4. What does the Mean Absolute Error (MAE) measure? **Answer:** It measures the average absolute difference between predicted and actual values in regression models.

5. What is a decision tree in predictive analytics?

   Answer: It's a model that uses a tree-like structure to make decisions or predictions based on input data features.

Check Your Progress 3:

1. Why is "correlation is not causation" an important warning in data analysis?
   Answer: Because two variables moving together does not imply that one causes the other; a third variable or coincidence might be the reason.

2. What is Simpson's Paradox?

   Answer: It occurs when a trend observed in separate groups reverses when the groups are combined, often due to confounding variables.

3. What is data dredging (p-hacking)?

   Answer: It refers to performing excessive tests on data to find statistically significant results without pre-defined hypotheses, often leading to false positives.

4. Give an example of a misleading correlation.

   Answer: A correlation between ice cream sales and drowning incidents— both increase in summer but one does not cause the other.

5. How can analysts avoid falling into the trap of data dredging? **Answer:** By predefining hypotheses, limiting the number of statistical tests, using corrections for multiple comparisons, and validating results with new data.